Tempico Software

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Tausand Electronics

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Tempico Software is a program developed by Tausand Electronics, with Joan Amaya and David Guzman as the lead developers. It is implemented in Python, utilizing key libraries such as PyQt5 (integrated with PySide2) for the graphical user interface, HIDAPI for serial communication, and PyGraph for data visualization. The software facilitates the use of Tausand Tempico TP1004 devices, streamlining the creation of histograms and lifetime measurements. For more information, visit tausand.com.

CHAPTER

ONE

STARTSTOPLOGIC MODULE

class Logic.StartStopLogic.**StartStopLogic**(*parent*, *disconnect*, *device*, *check1*, *check2*, *check3*, *check4*, *startbutton*, *stopbutton*, *savebutton*, *save_graph_1*, *clear_channel_A*, *clear_channel_B*, *clear_channel_C*, *clear_channel_D*, *connect*, *mainWindow*, *statusValue*, *statusPoint*, *timerStatus*, **args*, ***kwargs*)

Bases: `object`

This class handles the functionality for managing buttons, checkboxes, and graphs for histograms related to the Start-Stop measurements. It controls which channels are active, manages the display of histograms, and records the data from the Tempico device based on the user's interactions. It also handles the zooming feature for creating bars according to the time range received from the device.

Parameters

- **parent** – The parent QWindow for the logic.
- **disconnect** – The QPushButton used to disconnect.
- **device** – The Tempico device class that handles the measurements.
- **check4** (*check1, check2, check3,*) – QCheckBoxes that control the visibility of each histogram.
- **stopbutton** (*startbutton,*) – QPushButtons to start and stop the measurement process.
- **save_graph_1** (*savebutton,*) – QPushButtons to save data and save the graph.
- **clear_channel_D** (*clear_channel_A, clear_channel_B, clear_channel_C,*) – QPushButtons to clear data for each respective channel.
- **connect** – QPushButton to connect to the Tempico device.
- **mainWindow** – The main window (QWindow) for the GUI.
- **statusPoint** (*statusValue,*) – QLabel widgets for displaying status information (e.g., values and points).

autoRangeA()

Automatically adjusts the zoom level of channel A to the maximum value in the data.

The function sets a sentinel for zoom changes, checks if there are any data points in channel A, and if so, finds the maximum value. It then calls the changeZoomMax function to adjust the zoom accordingly.

Returns None

autoRangeB()

Automatically adjusts the zoom level of channel B to the maximum value in the data.

The function sets a sentinel for zoom changes, checks if there are any data points in channel B, and if so, finds the maximum value. It then calls the changeZoomMax function to adjust the zoom accordingly.

Returns None

`autoRangeC()`

Automatically adjusts the zoom level of channel C to the maximum value in the data.

The function sets a sentinel for zoom changes, checks if there are any data points in channel C, and if so, finds the maximum value. It then calls the changeZoomMax function to adjust the zoom accordingly.

Returns None

`autoRangeD()`

Automatically adjusts the zoom level of channel D to the maximum value in the data.

The function sets a sentinel for zoom changes, checks if there are any data points in channel D, and if so, finds the maximum value. It then calls the changeZoomMax function to adjust the zoom accordingly.

Returns None

`changeColorThread(color)`

`changeStatusColor(color)`

Changes the color of the status point in the status bar based on the input value.

This method updates the status point's color by drawing a filled circle with the specified color. The mapping of numerical values to colors is as follows: - 0: Gray - 1: Green - 2: Yellow - 3: Orange

Parameters `color` – A numerical value representing the desired color (int). 0 for gray, 1 for green, 2 for yellow, 3 for orange.

Returns None

`changeStatusThread(newText)`

Updates the text displayed in the status bar.

This method sets the status value text to the specified new text, reflecting the current status or information relevant to the user.

Parameters `newText` – The text to display in the status bar (str).

Returns None

`changeZoomMax(newMaxValue, channel)`

Automatically adjusts the zoom level of the plot to the maximum value found.

The function sets the x-axis range of the specified channel's plot to [0, newMaxValue]. It considers whether measurements have been taken and limits the zoom change based on the sentinel zoom change counters.

Parameters

- `newMaxValue` – The new maximum value for the zoom range (float).
- `channel` – The channel to which the zoom change applies (str). Expected values are 'A', 'B', 'C', or 'D'.

Returns None

`clear_a()`

Clears the lists containing measured data for channel A by resetting them to empty lists.

Returns None

`clear_b()`

Clears the lists containing measured data for channel B by resetting them to empty lists.

Returns None

`clear_c()`

Clears the lists containing measured data for channel C by resetting them to empty lists.

Returns None

`clear_d()`

Clears the lists containing measured data for channel D by resetting them to empty lists.

Returns None

`createDialog()`

Creates a dialog box to notify the user when the connection with the device has been lost.

This method displays a critical error message box indicating that the connection with the device has been lost. It also stops any ongoing graphic updates, hides relevant graphics, and disables the disconnect button while enabling the connect button.

Returns None

`create_graphs()`

Creates histograms and plots for the selected channels (A, B, C, D) and arranges them on the GUI.

This function initializes the graphical components and assigns the appropriate properties to the plots based on the selected channels. It also manages the layout of the graphs, resizing them according to the number of selected channels (1, 2, 3, or 4). After setting up the graphical components, the function starts a worker thread that handles the measurement process in parallel with the GUI thread, ensuring the UI remains responsive.

Returns None

`dialogChangeMode(channel)`

Opens a dialog box to ask the user if they want to change the mode for the selected channel.

The dialog informs the user that the collected data for the specified channel falls mostly outside the reliable range of mode 1 and prompts them to switch to mode 2. If the user confirms, the function updates the mode and processes the data accordingly.

Parameters `channel` – The channel for which to change the mode (str). Expected values are 'channel A', 'channel B', 'channel C', or 'channel D'.

Returns None

`disconnectedDevice()`

Disables device interaction controls when the device is disconnected.

This function updates the GUI to reflect that the device is no longer connected by disabling the disconnect and start buttons and enabling the connect button.

Returns None

`hide_graphic2()`

Disables the start and stop buttons in the graphical interface.

This function is typically used to prevent user interaction during certain operations.

Returns None

`resetSaveSentinels()`

`save_graphic()`

Saves the current data according to the selected format specified in the dialog box.

The function starts by retrieving the default histogram name and the current date, which is formatted to create a unique filename. It then initializes lists to hold filenames, data, settings, and column names for the saved files.

A dialog box is displayed for the user to select the desired file format (txt, csv, or dat) for saving the data. Once the user accepts the selection, the function checks if the selected format has already been saved before.

If the selected format has not been saved, it collects data from the various channels (A, B, C, D) if their corresponding sentinel variables indicate they should be saved. It retrieves the average cycles, mode, number of stops, stop edge, and stop mask for each channel to include in the settings. The function then attempts to save the collected data in the specified format.

If the save operation is successful, a message box is displayed, confirming the successful save and showing the folder path and file names. If an error occurs during the save process, an error message box is shown.

If the selected format has already been saved, a message box is displayed with the previous save information.

Returns None

`save_plots()`

Saves the current plots in the selected image format.

This method opens a dialog for the user to select an image format (PNG, TIFF, or JPG) and saves the plots for channels A, B, C, and D if their respective flags are set to True. The plots are saved with a timestamp in the specified format in the default folder path.

Returns None

`show_graphic(`*device_new*`)`

Sets the device and its channels, enabling the start button for measurement.

Parameters `device_new` – The new device to be set (DeviceType).

Returns None

`startTimerConnection()`

Starts the timer that periodically checks the device connection.

The timer runs every 500 milliseconds to monitor connectivity. This is typically called after a measurement finishes or when the application is idle.

Returns None

`start_graphic()`

Starts the graphical representation of the measurement based on the selected channels.

If no channels are selected, it prompts the user to select at least one channel before proceeding. It also disables certain buttons to prevent interaction during the measurement process.

Returns None

`stopTimerConnection()`

Stops the timer responsible for checking the device connection.

This function is typically called when a measurement begins to avoid interference during the acquisition process.

Returns None

`stop_graphic()`

Enables the functions to initiate a new measurement while disabling the buttons to clear graphs. It also updates the status bar text and color.

Returns None

threadComplete()

Handles the completion of the measurement thread by enabling the ability to switch between windows. It also calls the stop_graphic() function to update the UI accordingly.

Returns None

threadRunning(*status*)

Updates the thread creation status.

This method sets the threadCreatedSentinel attribute to indicate whether a thread has been created based on the provided status value.

Parameters **status** – The status indicating whether a thread has been created (int). Use 0 to indicate that the thread has been created, and 1 to indicate that it has not.

Returns None

updateBashDataPure(*valuesPureA*, *valuesPureB*, *valuesPureC*, *valuesPureD*)

updateBashSignal(*valueListA*, *valueListB*, *valueListC*, *valueListD*)

updateDataPure(*value*, *channel*)

Updates the raw data values for the specified channel.

This method appends the new raw value (in picoseconds) to the list of data for the specified channel (A, B, C, or D).

Parameters

- **value** – The raw data value to append to the data list (float or int). This value is in picoseconds.
- **channel** – The channel identifier for which the value is to be updated (str). Accepted values are "A", "B", "C", and "D".

Returns None

updateSignal(*value*, *channel*)

Updates the normalized data values for the specified channel and refreshes the corresponding histogram.

This method appends the new value to the list of normalized data for the specified channel (A, B, C, or D) and calls the update_histogram method to refresh the histogram with the updated data.

Parameters

- **value** – The normalized value to append to the data list (float or int).
- **channel** – The channel identifier for which the value is to be updated (str). Accepted values are "A", "B", "C", and "D".

Returns None

update_histogram(*data*, *curve*, *indexChannel*)

Calculates and updates the histogram for the specified channel based on the measured data.

The function first checks the selected channel (A, B, C, or D) and retrieves the current view range from the corresponding view box. It then creates bins for the histogram based on this range, using a total of 61 bins.

Finally, it computes the histogram of the provided data and updates the specified curve in the plot with the histogram values.

Parameters

- **data** – The measured data used to calculate the histogram (list).

- **curve** – The curve item from the plot in PyQt5 with PyQtGraph that corresponds to the channel (PlotCurveItem).
- **indexChannel** – The identifier of the channel for the histogram, which can be "A", "B", "C", or "D" (str).

Returns None

`zoom_changedA()`

Automatically calculates and updates the histogram for channel A when the user zooms in on the corresponding graph.

The function checks if the zoom was triggered by code or user action. If it was triggered by the user, it increments the sentinel for zoom changes. It then retrieves the current x-range from the view box for channel A and computes the bin width for the histogram.

Next, it creates 61 bins based on the x-range and calculates the histogram using the measured data. The histogram data is then used to update the curve item for channel A in the plot.

Param None

Returns None

`zoom_changedB()`

Automatically calculates and updates the histogram for channel B when the user zooms in on the corresponding graph.

The function checks if the zoom was triggered by code or user action. If it was triggered by the user, it increments the sentinel for zoom changes. It then retrieves the current x-range from the view box for channel B and computes the bin width for the histogram.

Next, it creates 61 bins based on the x-range and calculates the histogram using the measured data. The histogram data is then used to update the curve item for channel B in the plot.

Param None

Returns None

`zoom_changedC()`

Automatically calculates and updates the histogram for channel C when the user zooms in on the corresponding graph.

The function checks if the zoom was triggered by code or user action. If it was triggered by the user, it increments the sentinel for zoom changes. It then retrieves the current x-range from the view box for channel C and computes the bin width for the histogram.

Next, it creates 61 bins based on the x-range and calculates the histogram using the measured data. The histogram data is then used to update the curve item for channel C in the plot.

Param None

Returns None

`zoom_changedD()`

Automatically calculates and updates the histogram for channel D when the user zooms in on the corresponding graph.

The function checks if the zoom was triggered by code or user action. If it was triggered by the user, it increments the sentinel for zoom changes. It then retrieves the current x-range from the view box for channel D and computes the bin width for the histogram.

Next, it creates 61 bins based on the x-range and calculates the histogram using the measured data. The histogram data is then used to update the curve item for channel D in the plot.

Param None

Returns None

CHAPTER

TWO

G2LOGIC MODULE

class Logic.G2Logic.**G2Logic**(*stopChannelComboBox: PySide2.QtWidgets.QComboBox*, *coincidenceWindowComboBox: PySide2.QtWidgets.QComboBox*, *numberMeasurementsSpinBox: PySide2.QtWidgets.QSpinBox*, *numberBinsLabel: PySide2.QtWidgets.QLabel*, *startManualButton: PySide2.QtWidgets.QPushButton*, *stopManualButton: PySide2.QtWidgets.QPushButton*, *clearButton: PySide2.QtWidgets.QPushButton*, *saveDataButton: PySide2.QtWidgets.QPushButton*, *savePlotButton: PySide2.QtWidgets.QPushButton*, *comboBoxEquation: PySide2.QtWidgets.QComboBox*, *applyFitButton: PySide2.QtWidgets.QPushButton*, *parametersTable: PySide2.QtWidgets.QTableWidget*, *initialParametersButton: PySide2.QtWidgets.QPushButton*, *statusValueLabel: PySide2.QtWidgets.QLabel*, *statusColorLabel: PySide2.QtWidgets.QLabel*, *totalStartsLabel: PySide2.QtWidgets.QLabel*, *totalStopsLabel: PySide2.QtWidgets.QLabel*, *calculatedParameter: PySide2.QtWidgets.QLabel*, *helpButton: PySide2.QtWidgets.QPushButton*, *graphicFrame: PySide2.QtWidgets.QFrame*, *startLimitedButtonG2: PySide2.QtWidgets.QPushButton*, *stopLimitedButtonG2: PySide2.QtWidgets.QPushButton*, *clearLimitedButtonG2: PySide2.QtWidgets.QPushButton*, *autoClearSpinBox: PySide2.QtWidgets.QSpinBox*, *startAutoClearButton: PySide2.QtWidgets.QPushButton*, *stopAutoClearButton: PySide2.QtWidgets.QPushButton*, *clearAutoClearButton: PySide2.QtWidgets.QPushButton*, *maximumTimeComboBox: PySide2.QtWidgets.QComboBox*, *tabSettings: PySide2.QtWidgets.QTabWidget*, *fixedDelayCheckBox: PySide2.QtWidgets.QCheckBox*, *externalDelaySpinBox: PySide2.QtWidgets.QDoubleSpinBox*, *device: pyTempico.core.TempicoDevice*, *mainWindow*, *connectedTimer: PySide2.QtCore.QTimer*)

Bases: `object`

Manages the logic for the g^2 (second-order correlation) tab in the GUI.

This class is responsible for handling the interaction between the user interface and the underlying measurement logic for g^2 experiments. It connects UI elements, manages measurement control (manual, limited, and auto-clear modes), handles saving data/plots, and provides integration with fitting equations and parameter configuration.

Key responsibilities: - Control of manual, limited, and auto-clear measurements. - Management of fitting parameters and application of selected models. - Handling of measurement reset, saving, and plot export. - Connection of GUI elements (buttons, combo boxes, spin boxes, etc.)

to their respective actions.

- Preparation and update of the graphical frame for g^2 visualization.

  Parameters

  - **stopChannelComboBox** – Combo box to select the stop channel (QComboBox).
  - **coincidenceWindowComboBox** – Combo box to select the coincidence window (QComboBox).
  - **numberMeasurementsSpinBox** – Spin box to define the number of measurements (QSpinBox).
  - **numberBinsLabel** – Label to display the number of histogram bins (QLabel).
  - **startManualButton** – Button to start a manual measurement (QPushButton).
  - **stopManualButton** – Button to stop a manual measurement (QPushButton).
  - **clearButton** – Button to clear the current measurement (QPushButton).
  - **saveDataButton** – Button to save g^2 measurement data (QPushButton).
  - **savePlotButton** – Button to save the g^2 plot (QPushButton).
  - **comboBoxEquation** – Combo box to select the fitting equation (QComboBox).
  - **applyFitButton** – Button to apply the selected fit to the data (QPushButton).
  - **parametersTable** – Table to show and edit fitting parameters (QTableWidget).
  - **initialParametersButton** – Button to restore or configure initial fit parameters (QPushButton).
  - **statusValueLabel** – Label to display the current measurement status (QLabel).
  - **statusColorLabel** – Label to show the status with a color indicator (QLabel).
  - **totalStartsLabel** – Label to display the total number of start events (QLabel).
  - **totalStopsLabel** – Label to display the total number of stop events (QLabel).
  - **calculatedParameter** – Label to show the calculated parameter value (QLabel).
  - **helpButton** – Button to open the help dialog (QPushButton).
  - **graphicFrame** – Frame where the g^2 plot is displayed (QFrame).
  - **startLimitedButtonG2** – Button to start a limited measurement (QPushButton).
  - **stopLimitedButtonG2** – Button to stop a limited measurement (QPushButton).
  - **clearLimitedButtonG2** – Button to clear a limited measurement (QPushButton).
  - **autoClearSpinBox** – Spin box to configure auto-clear interval (QSpinBox).
  - **startAutoClearButton** – Button to start auto-clear measurements (QPushButton).
  - **stopAutoClearButton** – Button to stop auto-clear measurements (QPushButton).
  - **clearAutoClearButton** – Button to clear auto-clear measurement results (QPushButton).
  - **maximumTimeComboBox** – Combo box to select the maximum integration time (QComboBox).
  - **tabSettings** – Tab widget containing g^2 settings (QTabWidget).
  - **fixedDelayCheckBox** – Checkbox to enable/disable fixed delay (QCheckBox).
  - **externalDelaySpinBox** – Spin box to set the external delay value (QDoubleSpinBox).

- **device** – The Tempico device used for data acquisition (tempico.TempicoDevice).
- **mainWindow** – Reference to the main application window.
- **connectedTimer** – Timer used to monitor device connection (QTimer).

antiBunching(*t*, *T_c*)

Computes the antibunching correlation function.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_c** – Characteristic antibunching time (float)

Returns Antibunching function values as np.ndarray

antiBunchingShift(*t*, *T_c*, *T_d*, *b*)

Computes the shifted antibunching correlation function.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_c** – Characteristic antibunching time (float)
- **T_d** – Time shift applied to the function (float)
- **b** – Baseline offset added to the function (float)

Returns Shifted antibunching function values as np.ndarray

antiBunchingShiftFixed(*t*, *T_c*, *b*)

Computes the shifted antibunching correlation function with a fixed time shift.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_c** – Characteristic antibunching time (float)
- **b** – Baseline offset added to the function (float)

Returns Shifted antibunching function values as np.ndarray

applyFit(*fitName*)

Performs a curve fit for a selected equation using the measured g^2 data and initial parameter guesses, and returns the fitted parameter values along with their estimated standard deviations.

The method supports fits for thermal Gaussian, GaussianShift, thermal Lorentzian, LorentzianShift, AntiBunching, and AntiBunchingShift. Initial parameters are automatically adjusted according to the measurement units (ns or other) and the method can handle fits where certain delay parameters are fixed based on the externalDelayGaussianCheckBox.

When the fit is successful, the function returns the fitted parameters in the order required by the corresponding equation, along with their estimated variances or "N/A" for fixed parameters. If the fit fails due to any reason such as non-convergence, the method shows a dialog informing the user and returns empty lists for both parameters and standard deviations.

Parameters **fitName** – Name of the fit to apply (str). Must match one of the recognized fit types.

Returns A tuple containing the list of fitted parameter values and the list of corresponding standard deviations.

applyFitAction()

Applies the selected fit to the current data. The function disables the fit button and equation selector during processing. It determines which equation is selected, applies the corresponding fit, and updates the parameter table with the fit results. Once done, it re-enables the UI controls.

Returns None

applyFitToTable(*parametersList*, *stdList*, *fitApplied*)

Applies the results of a curve fit to the corresponding parameters in the application, updates the plotted curve, calculates the goodness of fit (R^2), and updates the parameter table in the UI. If the fit fails, produces NaN values, or the parameter list is empty, the method resets the parameters for the selected equation to their default state and notifies the user with a dialog.

The function handles six types of fits: - Thermal Gaussian - Gaussian Shift - Thermal Lorentzian - Lorentzian Shift - AntiBunching - AntiBunching Shift

For each fit type, it assigns the fitted parameter values and their standard deviations to internal variables, computes the fitted g^2 curve, and calculates R^2. If the fitted curve contains invalid values (NaNs), the method resets the parameters and shows a warning dialog. After updating or resetting the parameters, the corresponding table in the UI is refreshed to reflect the current values.

Parameters

- **parametersList** – List of fitted parameter values (list). The order depends on the selected fit type.
- **stdList** – List of standard deviations of the fitted parameters (list). Same order as parametersList.
- **fitApplied** – Name of the fit applied (str). Must be one of the recognized fit types.

Returns None

calculateR2(*data*, *fitData*)

Computes the coefficient of determination (R^2) between measured and fitted data.

Parameters

- **data** – Measured data points (list).
- **fitData** – Fitted data points (list).

Returns R^2 value rounded to two decimal places (float).

cancelParameters()

Cancels the parameter dialog.

This function checks the parameter values and closes the dialog without applying changes.

Returns None

captureMeasurement(*g2Values*, *totalStarts*, *totalStops*)

Captures the current g^2 measurement from the worker thread and updates the plot and labels.

This function stores the g^2 values, updates the main curve in the plot, and refreshes the total START and STOP counts displayed in the UI.

Parameters

- **g2Values** – List of g^2 values from the measurement (list).
- **totalStarts** – Total counts detected on the START channel (int).
- **totalStops** – Total counts detected on the STOP channel (int).

Returns None

captureTauValues(*tauValues*, *unitFactor*, *mode*)

Captures tau (time delay) values from the measurement thread and updates the G^2 plot.

This function sets the tauValues attribute, adjusts the plot's X and Y limits, and adds a non-movable shaded region indicating the measurement range. The region size depends on the acquisition mode and unit factor.

Parameters

- **tauValues** – List of tau values from the measurement (list of float).
- **unitFactor** – Conversion factor to adjust the measurement units (float).
- **mode** – Acquisition mode of the device, affecting the region size (int).

Returns None

changeAntiBunchingTableParameters()

Updates the parameter table for the Antibunching equation.

Two rows are inserted for the TauA parameter and the R^2 value. Each row displays the parameter name, formatted value, standard deviation, and units if available. If a parameter value is "nan", the table shows "Undefined" and leaves the other columns empty.

Returns None

changeAntibunchingShiftTableParameters()

Updates the parameter table for the Antibunching Shift equation.

Four rows are inserted for the parameters TauA, Taud, B, and R^2. Each row displays the parameter name, formatted value, standard deviation, and units if available. If a parameter value is "nan", the table shows "Undefined" and leaves the other columns empty.

Returns None

changeDeterminedParameters()

Sets the flag indicating that the measurement parameters have been determined.

Returns None

changeEstimatedParameter(*text*)

Updates the label showing the currently estimated parameter.

Parameters **text** – The new estimated parameter value to display (str).

Returns None

changeEstimatedParameterStartTimer(*text*)

Updates the label with the estimated parameter and starts the auto-clear timer. The timer will trigger at intervals defined by the user in the auto-clear spin box.

Parameters **text** – The new estimated parameter value to display (str).

Returns None

changeExternalCheckBox()

Updates the state of the main UI checkbox.

This function synchronizes the checkbox in the main interface with the stored variable value. Depending on the current index of the equation combo box, the main checkbox is updated to reflect the corresponding saved state.

Returns None

changeExternalFixed()

Updates the stored variable according to the main UI checkbox state.

This function sets the value of the corresponding variable based on whether the main checkbox is selected or not. The assignment depends on the current index of the equation combo box.

Returns None

`changeExternalSpinBox()`

Updates the main UI spin box with the correct units.

This function modifies the value, suffix, step size, and decimals of the main spin box (*externalDelaySpinBox*) depending on the selected equation in the combo box. The units are set to nanoseconds (ns) if the value is below 1000, or microseconds (us) otherwise. The corresponding stored initial delay value is used for the assignment.

Returns None

`changeReverseExternalSpinBox()`

Updates the internal delay variable based on the value set in the main spin box.

This function reads the current value and suffix ("ns" or "us") from *externalDelaySpinBox* and updates the corresponding internal variable (*thermalGaussianShiftTdInitial*). If the spin box value exceeds the unit threshold (>=1000 ns or <1 us), it automatically adjusts the units and rescales the displayed value. The update is applied according to the currently selected equation in the combo box.

Returns None

`changeStatus(text)`

Updates the status label with the provided text.

Parameters `text` – The new status text to display (str).

Returns None

`changeStatusColor(color)`

Changes the color of the point in the status bar. Each color is assigned a specific numeric value.

Parameters `color` – The numeric value corresponding to the desired color (int). - 0: Gray - 1: Green - 2: Yellow - 3: Orange

Returns None

`changeTableParameters()`

Updates the UI parameter table and the main curve plot according to the currently selected equation.

The method first updates the main UI elements, including the fixed delay checkbox and the external delay spinbox, to reflect the current state of the selected equation. It then clears the parameter table and fills it with the parameters corresponding to the selected equation type.

Depending on the selection in comboBoxEquation, it updates the table and the main plot with the corresponding fit data. If the fit data is empty, the main plot is cleared.

Returns None

`changeTermalGaussianShiftTableParameters()`

Updates the parameter table for the Thermal Gaussian Shifted equation.

This method inserts four rows in the parameters table. The first three rows correspond to the fit parameters Tc, Td, and B, and the fourth row shows the R^2 value of the fit.

For each parameter, if its value is not defined ("nan"), the table shows "Undefined" and leaves the remaining cells empty. If a value exists, it displays the formatted value, its standard deviation, and units. The R^2 row only displays the numeric value if available.

Returns None

changeTermalGaussianTableParameters()

Updates the parameter table specifically for the Thermal Gaussian equation.

This method inserts two rows into the parameters table. The first row corresponds to the Gaussian time constant (Tc) and the second row corresponds to the R^2 value of the fit.

If the value of Tc or R^2 is not available ("nan"), the table shows "Undefined" and leaves the other cells empty. Otherwise, it formats and displays the value, its standard deviation (if applicable), and the units for Tc, and the numeric value for R^2.

Returns None

changeTermalLorentzianShiftTableParameters()

Updates the parameter table for the Thermal Lorentzian Shifted equation.

Four rows are inserted corresponding to the fit parameters T0, Td, B, and the R^2 value. Each row displays the parameter name, formatted value, standard deviation, and units if available. If a parameter value is "nan", the table shows "Undefined" and leaves the other columns empty.

Returns None

changeTermalLorentzianTableParameters()

Updates the parameter table for the Thermal Lorentzian equation.

This method inserts two rows in the table. The first row corresponds to the T0 fit parameter and the second row to the R^2 value. If a parameter value is "nan", it displays "Undefined" and leaves the other cells empty. Otherwise, it shows the formatted value, its standard deviation, and units. The R^2 row only shows the numeric value if available.

Returns None

checkParametersValue()

Checks whether all initial fit parameters are set to their default values.

This function verifies if all the initial parameters are either 1 or 1000. If so, it sets the *parametersChange* flag to False, indicating no user modifications. Otherwise, the flag is set to True.

Returns None

checkRangesMode(*channel*, *maximumTime*)

Determines the optimal measurement mode for the device based on the selected channel and the maximum measurement time. Some device channels support multiple modes, and the appropriate mode is chosen to ensure accurate measurements within the specified maximum time. The selected mode is stored in the instance variable *modeToMeasure*.

Parameters

- **channel** – The channel to be measured (int).
- **maximumTime** – The maximum measurement time in picoseconds (int).

Returns None

clearManualMeasurement()

Clears the data captured during a manual measurement by calling the worker thread.

Returns None

connectedDevice(*device*)

Handles actions after a device is connected.

Parameters **device** – The connected device object.

Returns None

createGraphic()

Creates the main g2(tau) plot using PyQtGraph.

This function sets up a GraphicsLayoutWidget inside the graphic frame, configures grid, axes labels, and a legend, and adds two curves: one for the experimental g2 data and another for the fitted curve. The plot is embedded into a QHBoxLayout for display in the UI.

Returns None

createGraphicToSave()

Creates a copy of the current g^2 plot for saving purposes without affecting the on-screen graph.

This function generates a new GraphicsLayoutWidget, copies the curves, labels, and legend from the main plot, and optionally adds a footer showing the current fit equation and parameters. The returned widget can be resized or modified independently of the main plot for saving.

Returns tuple containing - saveWinG2 (GraphicsLayoutWidget): The copied plot widget ready for saving. - savePlotG2 (PlotItem): The main plot area within the copied widget.

disconnectedDevice()

Handles actions after the device is disconnected.

Returns None

finishAutoClearMeasurement()

Finalizes an auto-clear measurement after the measurement thread completes. Restores the general settings for the UI, disables the stop button, enables the start button, disables the clear button, and re-enables the spin box for selecting the auto-clear interval.

Returns None

finishLimitedMeasurement()

Finalizes a limited measurement once the measurement thread finishes. Restores the general settings for the UI, disables the stop button, enables the start button, disables the clear button, and re-enables the spin box for selecting the number of measurements.

Returns None

finishManualMeasurement()

Finalizes a manual measurement once the measurement thread finishes. Restores the general settings for the UI, disables the stop button, enables the start button, and disables the clear button.

Returns None

formatStd(*valueVar*, *valueEstimated*)

Formats the standard deviation of a parameter for display in the table.

If the standard deviation is "N/A", it returns it unchanged. Negative values return "nan". If the relative uncertainty (std / estimated value) exceeds 3, it returns "nan". Values of 0 are formatted as "0.00". Values between 0.01 and 1,000,000 are shown with two decimals, and other values are displayed in scientific notation with two significant figures.

Parameters

- **valueVar** – The variance or standard deviation to format (float, int, or "N/A").
- **valueEstimated** – The estimated value of the parameter (float or int) used to calculate relative uncertainty.

Returns Formatted string representing the standard deviation (str).

formatValue(*value*)

Formats a numerical value for display in the parameter table.

Values between 0.01 and 1,000,000 are shown with two decimal places. Values outside this range are displayed in scientific notation with two significant figures. Zero is formatted as "0.00".

Parameters `value` – The numerical value to format (float or int).

Returns Formatted string representing the value (str).

`generalSettingsAfterMeasurement`(*channel*)

Restores general UI and measurement settings after a measurement ends. This includes enabling/disabling buttons and combo boxes, setting initial parameters if tau values were captured, updating the status and finish time, and re-enabling the appropriate tabs depending on which measurement mode was active.

Parameters `channel` – Index of the measurement mode that was active (0: Manual, 1: Limited, 2: AutoClear) (int)

Returns None

`generalSettingsBeforeMeasurement`(*tab*)

Applies general pre-measurement settings for all three types of measurements (manual, limited, auto-clear). It stops any running timers, disables controls that should not be changed during measurement, resets data arrays, flags, and fit parameters. Also manages which tabs are enabled based on the selected measurement type.

Parameters `tab` (*int*) – Index indicating the type of measurement (0: manual, 1: limited, 2: auto-clear).

Returns None

`getChannelComboBox`()

Returns the letter of the currently selected STOP channel from the combo box.

Returns The channel letter as a string ("A", "B", "C", or "D").

`getPicoSecondsValue`(*valueStr*)

Converts a time value with units (ps, ns, us, ms) to picoseconds.

Parameters `valueStr` – A string containing the value and its unit, e.g., '10 ns' (str).

Returns The value converted to picoseconds (float).

`getUnits`()

Retrieves the time units selected in the maximum time combo box, updates the internal *unitsMeasured* variable, updates the plot label and parameter units accordingly, and returns the selected units.

Returns The selected time units as a string ('ps', 'ns', 'us', or 'ms').

`helpEventFilter`(*obj*, *event*)

Global event filter for the help '?' button.

Parameters

- `dialog` – QDialog being filtered.
- `obj` – QObject that received the event.
- `event` – QEvent instance.

Returns True if handled, False otherwise.

`initLocalDialogParameters`()

Initializes the local dialog parameters based on the current stored values.

This function copies the internal state variables that hold the initial parameters for Gaussian, Lorentzian, and Antibunching models (including their shifted versions) into the corresponding local variables used by the parameter dialog. It also initializes the local checkbox states for external delay options.

Returns None

`initParametersEquationAntiBunching()`
Initializes the fit parameters for the Antibunching equation. All parameter values, standard deviations, units, and R^2 are set to default (not-a-number) values.

Returns None

`initParametersEquationAntiBunchingShift()`
Initializes the fit parameters for the shifted Antibunching equation. All values, standard deviations, units, and R^2 are set to default (not-a-number) values.

Returns None

`initParametersEquationThermalGaussian()`
Initializes the fit parameters for the Thermal Gaussian equation. Sets the parameter values, standard deviations, units, and R^2 to default (not-a-number) values.

Returns None

`initParametersEquationThermalGaussianShift()`
Initializes the fit parameters for the Thermal Gaussian Shifted equation. Sets the parameter values, standard deviations, units, and R^2 to default (not-a-number) values.

Returns None

`initParametersEquationThermalLorentzian()`
Initializes the fit parameters for the Thermal Lorentzian equation. Sets the parameter value, standard deviation, units, and R^2 to default (not-a-number) values.

Returns None

`initParametersEquationThermalLorentzianShift()`
Initializes the fit parameters for the shifted Thermal Lorentzian equation. All parameter values, standard deviations, units, and R^2 are set to default (not-a-number) values.

Returns None

`initalParametersAntiBunching()`
Initializes the default parameter for the Anti-Bunching model.

Returns None

`initalParametersAntiBunchingShifted()`
Initializes the default parameters for the shifted Anti-Bunching model.

Returns None

`initalParametersLorentzian()`
Initializes the default parameter for the Lorentzian thermal model.

Returns None

`initialConfigs()`
Initializes the default configuration for the application interface and fitting parameters.

This method disables buttons related to starting, stopping, clearing, and saving measurements when the device is not connected. It also sets default suffixes, initializes fitting parameter equations, updates tables, and prepares the interface for user interaction.

Returns None

`initialExternalDelayCheckbox()`
Resets the external delay checkboxes for Gaussian, Lorentzian, and Anti-Bunching models to False.

Returns None

`initialParameters()`
Initializes all default fitting parameters by calling the corresponding setup methods for Gaussian, Lorentzian, and Anti-Bunching models.

Returns None

`initialParametersGaussian()`
Initializes the default parameter for the Gaussian thermal model.

Returns None

`initialParametersLorentzianShifted()`
Initializes the default parameters for the shifted Lorentzian thermal model.

Returns None

`initialParametersWithUnits()`
Initializes the default parameters for all models depending on the measurement units.

If parameters have not been changed manually, the function sets values close to 1 when the units are nanoseconds ("ns"). For other units, the parameters are initialized with values around 1000. This initialization applies to Gaussian, shifted Gaussian, Lorentzian, shifted Lorentzian, Anti-Bunching, and shifted Anti-Bunching models.

Returns None

`initialParametesGaussianShifted()`
Initializes the default parameters for the shifted Gaussian thermal model.

Returns None

`parametersFitName()`
Initializes the parameter names used in curve fitting.

Returns None

`resetFits()`
Resets all fit data and related parameters for every type of measurement. Clears the calculated fit arrays, reinitializes the fit parameters, updates the parameters table, clears the displayed curve, and resets all save sentinels.

Returns None

`resetSaveSentinels()`
Resets the sentinel values for saving files.

Returns None

`saveG2Data()`
Saves the current g^2 measurement data along with associated fit parameters and channel settings.

This function opens a dialog for the user to select the desired text format (txt, csv, dat). It checks if the data has already been saved in that format to avoid duplicates. The g^2 values and tau values are rounded for precision, and any calculated fit parameters for Gaussian, Gaussian Shifted, Lorentzian, Lorentzian Shifted, Antibunching, and Antibunching Shifted models are included in the saved file with their corresponding units. The save location and filename are generated based on the current date and a predefined prefix. After saving, a message box informs the user of the file path and name or reports an error if the save was unsuccessful.

Returns None

saveG2Plot()
Saves the graph image in the specified format (PNG, TIFF, or JPG) based on the user's selection. :return: None

setDefaultParameters()
Sets the default parameters for the selected model.

This function initializes the parameters depending on the measurement units (ns or ps) and updates the spin boxes in the dialog with the corresponding default values.

Returns None

setVerticalLabel()
Updates the Y-axis label of the g2(tau) plot based on the selected stop channel.

The label reflects which channel (A, B, C, or D) is currently used as the stop channel in the correlation measurement.

Returns None

showDialogChangeMode(*channel*)

showDialogNoFitParameters()
Displays a warning dialog indicating that the fit parameters could not be estimated.

Returns None

showDialogNoParameters()
Displays a warning dialog indicating that there are insufficient measurements on the START or STOP channel, making it impossible to determine the parameters required to calculate g^2.

Returns None

showParameterDialog()
Shows the dialog window for configuring initial parameters.

This function creates and displays a dialog that allows the user to configure initial parameters for Gaussian, Lorentzian, and Antibunching models (including shifted versions). The dialog contains a combo box for selecting the model, dynamic parameter fields, and control buttons (Apply, Default settings, Cancel). It also integrates a help button that opens the parameter help window.

Returns A tuple (bool, list). The boolean indicates whether the dialog was accepted (True) or canceled (False). The list contains the field widgets created for the selected parameters.

showParameterHelp()
Shows a fixed message.

Returns None

startAutoClearMeasurement()
Starts an auto-clear measurement where the plot is periodically cleared based on the user-selected time interval. This measurement continuously collects data while automatically resetting the graph at each interval.

Sets up and starts the worker thread for data collection and a QTimer to trigger the clearing of the graph.

Returns None

startLimitedMeasurement()
Starts a limited manual measurement in which the user specifies the number of measurements to perform. After reaching this number, the measurement automatically stops.

Configures the measurement parameters, disables/enables the corresponding buttons and spinboxes, determines the device mode based on the selected channel and maximum time, and starts the worker thread.

Returns None

`startManualMeasurement()`

Starts a manual measurement session. The user can start and stop the measurement using the corresponding buttons. Initializes measurement settings, retrieves selected parameters, and starts the worker thread for data acquisition.

Returns None

`startTimerConnection()`

Starts the connection timer with a 500 ms interval.

Returns None

`stopAutoClearMeasurement()`

Stops the auto-clear measurement. This stops the data collection worker thread and disables the QTimer that periodically clears the graph.

Returns None

`stopLimitedMeasurement()`

Stops the ongoing limited manual measurement and disables the corresponding stop button.

This interrupts the worker thread responsible for collecting the data.

Returns None

`stopManualMeasurement()`

Stops the ongoing manual measurement by stopping the worker thread and disables the stop button.

Returns None

`stopTimerConnection()`

Stops the connection timer.

Returns None

`thermalGaussian(t, T_c)`

Computes the thermal Gaussian correlation function.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_c** – Correlation time (float)

Returns Values of the thermal Gaussian function at each time t (np.ndarray)

`thermalGaussianShift(t, T_c, T_d, b)`

Computes the thermal Gaussian correlation function with a time shift and baseline.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_c** – Correlation time (float)
- **T_d** – Time shift (float)
- **b** – Baseline offset (float)

Returns Values of the shifted thermal Gaussian function at each time t (np.ndarray)

`thermalGaussianShiftFixed(t, T_c, b)`

Computes the thermal Gaussian correlation function with a fixed time shift.

The time shift T_d is taken from the initial parameter depending on the units measured.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_c** – Correlation time (float)
- **b** – Baseline offset (float)

Returns Values of the shifted thermal Gaussian function at each time t (np.ndarray)

thermalLorentzian(*t*, *T_0*)

Computes the thermal Lorentzian correlation function.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_0** – Characteristic decay time (float)

Returns Values of the thermal Lorentzian function at each time t (np.ndarray)

thermalLorentzianShift(*t*, *T_0*, *T_d*, *b*)

Computes the thermal Lorentzian correlation function with a time shift and offset.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_0** – Characteristic decay time (float)
- **T_d** – Time shift (float)
- **b** – Constant offset (float)

Returns Values of the shifted thermal Lorentzian function at each time t (np.ndarray)

thermalLorentzianShiftFixed(*t*, *T_0*, *b*)

Computes the thermal Lorentzian correlation function with a fixed time shift and offset.

The time shift T_d is taken from the initial value stored in the object, and automatically converted if the units are not in nanoseconds.

Parameters

- **t** – Time values (list or np.ndarray)
- **T_0** – Characteristic decay time (float)
- **b** – Constant offset (float)

Returns Values of the thermal Lorentzian function with fixed shift at each time t (np.ndarray)

threadSettingsBeforeMeasurement(*tab*)

Connects signals from the worker thread responsible for executing the measurement to the appropriate GUI update functions. Sets up handling for status updates, color changes, estimated parameters, tau values, and measurement data capture. Also records the start time of the measurement and connects the worker's finished signal to the corresponding finish function based on the type of measurement.

Parameters **tab** (*int*) – Index indicating the type of measurement (0: manual, 1: limited, 2: auto-clear).

Returns None

updateInitialParameters()

Applies and saves the changes made in the Initial Parameters dialog.

This function updates the global initial parameters with the values set in the dialog's local variables for each type of equation: Gaussian, Gaussian shifted, Lorentzian, Lorentzian shifted, Antibunching, and Antibunching shifted. It also updates the state of the external delay checkboxes in the main UI to reflect whether the delay is fixed or not, and updates the spin box that stores the T_d value according to the current units. Finally, it calls checkParametersValue() to verify if the current values differ from the defaults and closes the dialog with accept().

Returns None

`updateLocalDialogInitalParameters()`

Updates the local copies of initial parameters based on the last selected fitting model and the values entered in the dialog spin boxes.

Depending on the selected model (Gaussian, shifted Gaussian, Lorentzian, shifted Lorentzian, Anti-Bunching, or shifted Anti-Bunching), the function reads the corresponding spin box values, applies unit conversion if needed (from microseconds to nanoseconds), and assigns them to the appropriate local variables. It also stores the fixed-state option from the checkbox when available.

Returns None

`updateParameterFields(`*comboBox*, *fieldLayout*, *fieldWidgets*, *dialog*`)`

Updates the dialog fields for the initial parameters.

This function rebuilds the parameter input fields in the dialog based on the current selection of the combo box. It updates the corresponding local variables with the current values, clears the previous layout, and dynamically adds the spin boxes and checkboxes required for the selected equation. The created widgets are connected to functions so that values are correctly saved when modified.

Parameters

- **`comboBox`** – Combo box containing the equation selection (QComboBox).
- **`fieldLayout`** – Layout where the parameter fields will be placed (QLayout).
- **`fieldWidgets`** – List that stores the generated field widgets (list).
- **`dialog`** – Dialog window containing the parameter fields (QDialog).

Returns None

`updateParameterUnits(`*unitsLabel*`)`

Updates the units for all initial measurement parameters to the specified label.

Parameters **`unitsLabel`** – The units to assign to the parameters (str), e.g., 'ps', 'ns', 'us', or 'ms'.

Returns None

`updateSpinBoxSufix(`*spinbox*, *value*`)`

Updates the QDoubleSpinBox value and suffix according to its magnitude. Converts between ns and us automatically and adjusts decimals and step.

Parameters

- **`spinbox`** – The QDoubleSpinBox being updated (QDoubleSpinBox)
- **`value`** – The new value of the spinbox (float)

Returns None

CHAPTER

THREE

LIFETIMELOGIC MODULE

class Logic.LifeTimeLogic.**LifeTimeLogic**(*comboBoxStartChannel: PySide2.QtWidgets.QComboBox*, *comboBoxStopChannel: PySide2.QtWidgets.QComboBox*, *graphicFrame: PySide2.QtWidgets.QFrame*, *startButton: PySide2.QtWidgets.QPushButton*, *stopButton: PySide2.QtWidgets.QPushButton*, *initialParametersButton: PySide2.QtWidgets.QPushButton*, *clearButton: PySide2.QtWidgets.QPushButton*, *saveDataButton: PySide2.QtWidgets.QPushButton*, *savePlotButton: PySide2.QtWidgets.QPushButton*, *statusLabel: PySide2.QtWidgets.QLabel*, *pointLabel: PySide2.QtWidgets.QLabel*, *binWidthComboBox: PySide2.QtWidgets.QComboBox*, *numberBins: PySide2.QtWidgets.QComboBox*, *functionComboBox: PySide2.QtWidgets.QComboBox*, *numberMeasurementsSpinBox: PySide2.QtWidgets.QSpinBox*, *totalMeasurements: PySide2.QtWidgets.QLabel*, *totalStart: PySide2.QtWidgets.QLabel*, *totalTime: PySide2.QtWidgets.QLabel*, *timeRange: PySide2.QtWidgets.QLabel*, *device*, *applyButton: PySide2.QtWidgets.QPushButton*, *parameterTable: PySide2.QtWidgets.QTableWidget*, *MainWindow*, *timerConnection: PySide2.QtCore.QTimer*)

Bases: `object`

Class responsible for the logic and functionality of the LifeTime (Fluorescence Lifetime Measurement) window.

This class manages the creation and updating of graphical plots related to LifeTime, controls user interactions with various buttons, and ensures that measurements are taken from the specified channels. The class handles: - Initialization of the graphical interface elements. - Starting and stopping of measurements. - Saving of data and plots. - Displaying parameters of the measurements, such as the number of measurements taken, and the percentage of completion.

Parameters

- **comboBoxStartChannel** – The combo box for selecting the start channel (QComboBox).
- **comboBoxStopChannel** – The combo box for selecting the stop channel (QComboBox).
- **graphicFrame** – The frame that holds the graphical plot (QFrame).
- **startButton** – The button to start the measurements (QPushButton).
- **stopButton** – The button to stop the measurements (QPushButton).
- **initialParametersButton** – The button to configure initial parameters (QPushButton).

- **clearButton** – The button to clear the graphs and data (QPushButton).
- **saveDataButton** – The button to save the measurement data (QPushButton).
- **savePlotButton** – The button to save the graph plot (QPushButton).
- **statusLabel** – The label showing the current status (QLabel).
- **pointLabel** – The label displaying the current point in the measurement (QLabel).
- **binWidthComboBox** – The combo box for selecting the bin width (QComboBox).
- **numberBins** – The combo box for selecting the number of bins (QComboBox).
- **functionComboBox** – The combo box for selecting a function (QComboBox).
- **numberMeasurementsSpinBox** – The spin box for specifying the number of measurements (QSpinBox).
- **totalMeasurements** – The label showing the total number of measurements (QLabel).
- **totalStart** – The label showing the number of measurements taken from the start channel (QLabel).
- **totalTime** – The label showing the total time of measurements (QLabel).
- **timeRange** – The label showing the time range for the measurements (QLabel).
- **device** – The connected Tempico device used for measurements.
- **applyButton** – The button for applying selected parameters (QPushButton).
- **parameterTable** – The table displaying the parameters of the measurements (QTableWidget).
- **MainWindow** – The main window that holds the application UI elements.
- **timerStatus** – The timer used for periodic updates (QTimer).

applyAction()

Captures the action of the 'apply' button, calculates the selected fit, and stores the resulting parameters.

This function determines which fitting function is selected from *functionComboBox* and attempts to calculate the fit using the measured time and data. If the parameters can be determined, they are stored in *FitParameters*. If not, a dialog informs the user that the parameters could not be determined.

Variables assignment: - The parameters *i0*, *tau0*, and other fit-specific variables are calculated using the corresponding fit function. - If the fit is successful, the rounded parameters are stored in *FitParameters*. - If the fit fails or returns "Undefined", a message box is shown to inform the user of the error.

Returns None

applyInitialDialog()

Updates the initial parameters based on user input from the parameter dialog.

This function is called when the user clicks the "Apply" button in the parameter dialog. It retrieves the values from the input fields corresponding to the selected fitting function and assigns them to the appropriate instance variables.

The following parameters are updated based on the selected fit type: - For Exponential fit: - I0 (initial value) - tau0 (decay constant) - For Kohlrausch fit: - I0 (initial value) - tau0 (decay constant) - beta (exponent) - For Shifted Exponential fit: - I0 (initial value) - tau0 (decay constant) - alpha (shift parameter) - b (baseline value) - For Double Exponential fit: - I0 (initial value) - tau0 (first decay constant) - tau1 (second decay constant) - alpha (mixing coefficient)

Variables assignment: - The updated values are assigned to their respective initial parameter variables. - A boolean flag indicating that initial parameters have changed is set to True for the relevant fit type.

Returns None

`calculateR2`(*data*, *fitData*)

Calculates the R^2 value based on the observed and fitted data.

Parameters

- `data` – The observed data points (array-like).
- `fitData` – The fitted data points (array-like).

Returns The R^2 value indicating the goodness of fit (float).

`changeCreatedStatus`()

Notifies that the measurement thread has been created.

This function sets the status of thread creation to True.

Returns None

`changeFunction`()

Updates the parameter values in the UI based on the selected function in the functionComboBox.

This function checks the current fitting function (e.g., "ExpDecay", "Kohlrausch", "ShiftedExponential", or "DoubleExponential") and updates the parameter labels accordingly. If the parameters have already been calculated, it will display the values along with their uncertainties and units. If the parameters are not available, it sets the values to "Undefined".

Variables assignment: - Parameters like "I0", "tau0", "Beta", "alpha", "b", "R^2", and their respective uncertainties are assigned using *insertParameters()* based on the fitting model. - Units are assigned through the *self.units* attribute for certain parameters (e.g., tau0).

Behavior by function: - ExpDecay: Updates *I0*, *tau0*, and *R^2*. - Kohlrausch: Updates *I0*, *tau0*, *Beta*, and *R^2*. - ShiftedExponential: Updates *I0*, *tau0*, *alpha*, *b*, and *R^2*. - DoubleExponential: Updates *I0*, *tau0*, *tau1*, *alpha*, and *R^2*.

Returns None

`changeStatusColor`(*color*)

Changes the color of the point in the status bar. Each color is assigned a specific numeric value.

Parameters `color` – The numeric value corresponding to the desired color (int). - 0: Gray - 1: Green - 2: Yellow - 3: Orange

Returns None

`changeStatusLabel`(*textValue*)

Changes the text of the status label.

Parameters `textValue` – The text to set for the status label (str).

Returns None

`checkDeviceStatus`()

Verifies the device's operational status by attempting to read its identification.

This function performs the following actions: - Attempts to read the identification from the device to ensure it is functioning correctly. - If the read operation fails, it stops any running worker thread and disconnects the device from the serial port. - Displays an error message indicating that the connection with the device has been lost.

Returns None

clearGraphic()

Clears the measured data and time lists when a measurement is in progress and also clears the graphic by executing the worker's clear function.

Returns None

connectedDevice(*device*)

Configures the UI options when the Tempico device is connected.

This function enables and disables the appropriate buttons and starts the timer to monitor the device status. - Enables the disconnect button and disables the connect button. - Starts a status timer with a 500 ms interval. - Assigns the connected device to the class attribute and enables the start button.

Parameters **device** – The connected Tempico device.

Returns None

disconnectedDevice()

Configures the UI options when the Tempico device is disconnected.

This function disables and enables the appropriate buttons and stops the timer that monitors the device status. - Disables the disconnect button and enables the connect button. - Stops the status timer. - Disables the start button.

Returns None

double_Exponential(*t*, *I0*, *tau0*, *tau1*, *alpha*)

Computes the value of a double exponential decay function.

This function calculates the value of a double exponential decay function using the given parameters: time *t*, initial value *I0*, decay constants *tau0* and *tau1*, and mixing parameter *alpha*.

Variables assignment: - The computed value is returned directly and not assigned to any variable within the function.

Parameters

- **t** – The time at which to evaluate the function (float).
- **I0** – The initial value of the exponential function (float).
- **tau0** – The first decay constant (float).
- **tau1** – The second decay constant (float).
- **alpha** – The mixing parameter that determines the contribution of the first decay (float).

Returns The value of the double exponential decay function at time *t*.

Return type float

enableAfterFinisihThread()

Executes when a measurement ends, either by pressing the stop button or when it has naturally finished. Re-enables the buttons, combo boxes, and fields necessary for a new measurement or for saving data. Resets the default parameters for the settings.

Returns None

exp_decay(*t*, *I0*, *tau0*)

Computes the value of an exponential decay function.

This function calculates the value of the exponential decay function using the given parameters for time *t*, initial value *I0*, and decay constant *tau0*.

Variables assignment: - The computed value is returned directly and not assigned to any variable within the function.

Parameters

- **t** – The time at which to evaluate the function (float).
- **I0** – The initial value of the exponential function (float).
- **tau0** – The decay constant (float).

Returns The value of the exponential decay function at time *t*.

Return type float

finishedThreadMeasurement()

Executes when the measurement thread finishes.

This function enables the UI components related to new measurements and resets the settings to default.

Returns None

fitDoubleExponential(*xData*, *yData*)

Fits a double exponential decay model to the provided data.

This function fits a double exponential decay model to the given *xData* and *yData* using the initial guesses for the parameters (I0, tau0, tau1, alpha). If the fit is successful, it updates the corresponding values for each parameter and plots the fitted curve. If the fit fails, a warning message is displayed to the user.

Variables assignment: - *self.FitCov*: Stores the covariance values for I0, tau0, tau1, and alpha, or "nan" if not calculable. - *self.xDataFitCopy*: Stores a copy of the *xData* for the fitted curve. - *self.yDataFitCopy*: Stores the computed y-values for the fitted curve. - *self.R2*: Stores the calculated R^2 value for the fit. - *self.curveFit*: Updates the plot with the fitted curve data.

Parameters

- **xData** – The x-axis data points (array-like).
- **yData** – The y-axis data points (array-like).

Returns Tuple (I0_opt, tau0_opt, tau1_opt, alpha_opt) containing the fitted values of I0, tau0, tau1, and alpha, or ("Undefined", "Undefined", "Undefined", "Undefined") if fitting fails.

Return type tuple(float, float, float, float) or tuple(str, str, str, str)

fitExpDecay(*xData*, *yData*)

Calculates the exponential decay fit for the given data.

This function attempts to fit an exponential decay model to the provided *xData* and *yData*. If successful, the optimal parameters (I0, tau0) are used to update the respective labels in the parameters table along with their uncertainties (covariance), and the fitted curve is plotted. If the fit cannot be performed, it alerts the user via a message box.

Variables assignment: - *self.FitCov*: Contains the covariance values for I0 and tau0, or "nan" if not calculable. - *self.xDataFitCopy*: Stores a copy of the *xData* for the fitted curve. - *self.yDataFitCopy*: Stores the computed y-values for the fitted curve. - *self.R2*: Stores the calculated R^2 value for the fit. - *self.curveFit*: Updates the plot with the fitted curve data.

Parameters

- **xData** – The x-axis data points (array-like).
- **yData** – The y-axis data points (array-like).

Returns Tuple (I0_opt, tau0_opt) containing the fitted values of I0 and tau0, or ("Undefined", "Undefined") if fitting fails.

Return type tuple(float, float) or tuple(str, str)

fitKohlrauschFit(*xData*, *yData*)

Calculates the Kohlrausch fit for the given data.

This function attempts to fit a Kohlrausch (stretched exponential) model to the provided *xData* and *yData*. If successful, the optimal parameters (I0, tau0, Beta) are used to update the respective labels in the parameters table along with their uncertainties (covariance), and the fitted curve is plotted. If the fit cannot be performed, it alerts the user via a message box.

Variables assignment: - *self.FitCov*: Contains the covariance values for I0, tau0, and Beta, or "nan" if not calculable. - *self.xDataFitCopy*: Stores a copy of the *xData* for the fitted curve. - *self.yDataFitCopy*: Stores the computed y-values for the fitted curve. - *self.R2*: Stores the calculated R^2 value for the fit. - *self.curveFit*: Updates the plot with the fitted curve data.

Parameters

- **xData** – The x-axis data points (array-like).
- **yData** – The y-axis data points (array-like).

Returns Tuple (I0_opt, tau0_opt, beta_opt) containing the fitted values of I0, tau0, and Beta, or ("Undefined", "Undefined", "Undefined") if fitting fails.

Return type tuple(float, float, float) or tuple(str, str, str)

fitShiiftedExponential(*xData*, *yData*)

Fits a shifted exponential decay model to the provided data.

This function fits a shifted exponential decay model to the given *xData* and *yData* using the initial guesses for the parameters (I0, tau0, alpha, b). If the fit is successful, it updates the corresponding values for each parameter and plots the fitted curve. If the fit fails, a warning message is displayed to the user.

Variables assignment: - *self.FitCov*: Stores the covariance values for I0, tau0, alpha, and b, or "nan" if not calculable. - *self.xDataFitCopy*: Stores a copy of the *xData* for the fitted curve. - *self.yDataFitCopy*: Stores the computed y-values for the fitted curve. - *self.R2*: Stores the calculated R^2 value for the fit. - *self.curveFit*: Updates the plot with the fitted curve data.

Parameters

- **xData** – The x-axis data points (array-like).
- **yData** – The y-axis data points (array-like).

Returns Tuple (I0_opt, tau0_opt, alpha_opt, b_opt) containing the fitted values of I0, tau0, alpha, and b, or ("Undefined", "Undefined", "Undefined", "Undefined") if fitting fails.

Return type tuple(float, float, float, float) or tuple(str, str, str, str)

getTempicoChannel()

Assigns the Tempico device channels based on the selected values from the combo boxes. The selected start and stop channels are assigned to the variables *currentStartChannel* and *currentStopChannel*, respectively. It also enables the corresponding channels.

Returns None

getUnits(*value*)

Receives a numerical value in picoseconds (ps) and returns a list with two values: the appropriate units for the value and a number indicating by how much to divide the value for conversion.

Parameters **value** – The numerical value in picoseconds to convert.

Returns A list containing the appropriate unit as a string and a number indicating the divisor for conversion.

`indexChangeStartChannel()`

Verifies that the options of the start and stop combo boxes are not the same when the start combo box is changed. If they are the same, it reverts to the previously selected option.

Returns None

`indexChangeStopChannel()`

Verifies that the options of the start and stop combo boxes are not the same when the stop combo box is changed. If they are the same, it reverts to the previously selected option.

Returns None

`initialParametersDialog()`

Displays a dialog for selecting initial parameters for fitting functions.

This dialog allows the user to select a type of fitting function and input the corresponding parameters. The parameters that can be set include: - I0 (initial value) - tau0 (decay constant) - beta (for Kohlrausch fit) - alpha (for Shifted Exponential fit) - b (for Shifted Exponential fit)

The dialog contains a ComboBox to select the function type, and the input fields dynamically update based on the selected function. It includes "Apply" and "Reset" buttons to apply changes or revert to default values.

Variables assignment: - The input values from the dialog are assigned to respective attributes, but are not directly returned.

Returns None

`insertParameters(`*index*, *Parameter*, *Value*, *Cov*, *Units*`)`

Inserts the parameter values into the fitting table at the specified row.

This function updates the table used for displaying the fitted parameters by inserting the parameter name, its value, the covariance (uncertainty), and the units in the specified row of the table.

Variables assignment: - The parameter name is assigned to the first column of the specified row (*index*). - The value of the parameter is assigned to the second column. - The covariance of the parameter is assigned to the third column. - The units of the parameter are assigned to the fourth column.

Parameters

- **`index`** – The row index where the values will be inserted (int).
- **`Parameter`** – The name of the parameter to be displayed (str).
- **`Value`** – The value of the parameter to be displayed (str).
- **`Cov`** – The covariance (uncertainty) of the parameter (str).
- **`Units`** – The units of the parameter (str).

Returns None

`kohl_decay(`*t*, *I0*, *tau0*, *beta*`)`

Computes the value of a Kohlrausch (stretched exponential) decay function.

This function calculates the value of the stretched exponential (Kohlrausch) decay function using the given parameters: time *t*, initial value *I0*, decay constant *tau0*, and stretching parameter *beta*.

Variables assignment: - The computed value is returned directly and not assigned to any variable within the function.

Parameters

- **`t`** – The time at which to evaluate the function (float).
- **`I0`** – The initial value of the exponential function (float).

- **tau0** – The decay constant (float).
- **beta** – The stretching exponent, controlling the deviation from a simple exponential decay (float).

Returns The value of the stretched exponential decay function at time *t*.

Return type float

`lostConnection()`

`maxRound(string)`

Determines the number of decimal places needed for at least three significant figures.

Parameters **string** – The number as a string (str).

Returns Number of decimal places (int).

`resetInitialDialog()`

Resets the initial parameters in the dialog based on the measured data.

This function is called when the user clicks the "Default Values" button in the parameter dialog. It resets the initial parameters to sensible defaults derived from the measured data. The following resets occur based on the selected fitting function:

- **For Exponential fit:**
  - I0 is set to the maximum value of measured data.
  - tau0 is set to the mean of measured time.
- **For Kohlrausch fit:**
  - I0 is set to the maximum value of measured data.
  - tau0 is set to the mean of measured time.
  - beta is set to 1.0 (default value).
- **For Shifted Exponential fit:**
  - I0 is set to the maximum value of measured data.
  - tau0 is set to the mean of measured time.
  - alpha is set to 0 (default value).
  - b is set to 0 (default value).
- **For Double Exponential fit:**
  - I0 is set to the maximum value of measured data.
  - tau0 is set to the mean of measured time.
  - alpha is set to 0 (default value).
  - tau1 is set to the mean of measured time.

The corresponding input fields in the dialog are updated with these new values. Flags indicating that the initial parameters have not changed are also reset to False for the relevant fit type.

Returns None

`resetParametersLabels()`

Resets all the fit parameters to 'Undefined' and clears the current fit type.

This function sets the list *FitParameters* to contain "Undefined" values for all parameters and clears *currentFit*. Additionally, it resets the *R2* value to "Undefined". Depending on the selected fit type in *functionComboBox*, the respective parameter labels in the table are updated to reflect "Undefined" values.

Variables assignment: - *FitParameters* is reset to a list of "Undefined". - *currentFit* is cleared to an empty string. - *R2* is reset to "Undefined". - The *insertParameters* method is used to update the table with "Undefined" values for each parameter based on the selected function in *functionComboBox*.

Returns None

`resetSaveSentinels()`

`roundStringPCov(number)`

Converts a number to a string representation in scientific notation with two decimal places.

Parameters `number` – The input number (float).

Returns String representation in scientific notation (str).

`saveLifeTimeData()`

Saves the data in the selected format (TXT, CSV, or DAT) based on the user's choice.

Returns None

`saveMode()`

Saves the current modes of the channels before starting a measurement. The modes are stored in the variables *oldChannelA*, *oldChannelB*, *oldChannelC*, and *oldChannelD*, allowing for restoration to the original settings after the measurement.

Returns None

`savePlotLifeTime()`

Saves the graph image in the specified format (PNG, TIFF, or JPG) based on the user's selection. :return: None

`setOldMode()`

Restores the modes of the channels after a measurement is completed. The channels are set to their previous modes stored in *oldChannelA*, *oldChannelB*, *oldChannelC*, and *oldChannelD*.

Returns None

`shifted_decay_function(t, I0, tau0, alpha, b)`

Computes the value of a shifted exponential decay function.

This function calculates the value of a shifted exponential decay function using the given parameters: time *t*, initial value *I0*, decay constant *tau0*, shift parameter *alpha*, and constant offset *b*.

Variables assignment: - The computed value is returned directly and not assigned to any variable within the function.

Parameters

- `t` – The time at which to evaluate the function (float).
- `I0` – The initial value of the exponential function (float).
- `tau0` – The decay constant (float).
- `alpha` – The shift parameter, determining the horizontal shift of the decay function (float).
- `b` – The constant offset added to the decay function (float).

Returns The value of the shifted exponential decay function at time *t*.

Return type float

`startMeasurement()`

Initiates the measurement process by clearing previous data, resetting the UI, and starting a new measurement thread.

This function performs several actions to prepare for and start a new measurement: - Stops the status timer and disables/enables relevant buttons. - Clears previous graph data and resets parameter labels. - Saves the current channel modes and resets save parameters. - Updates the status label and color to indicate a running measurement. - Clears previously measured data and time, resetting the plot. - Retrieves the selected channels and initializes a new measurement thread with the selected parameters. - Connects signals from the worker thread to corresponding slots in the main thread for UI updates.

Returns None

`startTimer()`

Starts the timer that triggers the update_timer function every second.

This function initiates the timer for the ongoing measurement, allowing the update_timer function to be executed at one-second intervals.

Returns None

`startTimerConnection()`

Starts the timer to check the device connection status every 500 milliseconds.

This method initiates periodic checks to monitor if the connected device remains active and responsive. The interval is set to 500 ms.

Returns None

`stopMeasurement()`

Stops the measurement thread if it is created and executes the function enableAfterFinishedThread() if the thread is not created.

Returns None

`stopTimer()`

Stops the timer that triggers the update_timer function.

This function halts the timer responsible for updating the measurement time and sets the totalTime label to indicate that no measurement is currently running.

Returns None

`stopTimerConnection()`

Stops the timer responsible for checking the device connection status.

This method halts the periodic checks performed by *timerConnection* to avoid unnecessary polling or to pause the connection validation during inactive states.

Returns None

`timeRangeValue()`

Converts the value from the timeRange label to picoseconds (ps).

Returns float The value converted to picoseconds.

`updateLabel(units)`

Updates the label of the x-axis in the graph with the correct units for the data.

This function takes the units as a string and updates the x-axis label accordingly.

Parameters `units` – The units to be displayed on the x-axis label (str).

Returns None

updateLabels(*totalMeasurements*, *totalStarts*)

Updates the labels for total measurements and total starts with the given values.

This function sets the text of the totalMeasurements and totalStart labels according to the parameters provided.

Parameters

- **totalMeasurements** – The text to set for the total measurements label (str).
- **totalStarts** – The text to set for the total starts label (str).

Returns None

updateMeasurement(*listOfNewValues*, *domainMeasurement*)

Updates the measured data and time with new values and refreshes the graph.

This function receives a new list of values representing counts and their corresponding update times, assigning them to the variables measuredData and measuredTime, and updates the graph accordingly.

Parameters

- **listOfNewValues** – The new list of measurement values (list).
- **domainMeasurement** – The list of times corresponding to the measurement values (list).

Returns None

update_timer()

Updates the timer for the ongoing measurement by adding one second.

This function increments the time variable by one second and updates the totalTime label to reflect the new time.

Returns None

CHAPTER

FOUR

COUNTSESTIMATEDLOGIC MODULE

class Logic.CountsEstimatedLogic.**CountEstimatedLogic**(*channelACheckBox: PySide2.QtWidgets.QCheckBox, channelBCheckBox: PySide2.QtWidgets.QCheckBox, channelCCheckBox: PySide2.QtWidgets.QCheckBox, channelDCheckBox: PySide2.QtWidgets.QCheckBox, startButton: PySide2.QtWidgets.QPushButton, stopButton: PySide2.QtWidgets.QPushButton, mergeRadio: PySide2.QtWidgets.QRadioButton, separateGraphics: PySide2.QtWidgets.QRadioButton, deatachedGraphics: PySide2.QtWidgets.QRadioButton, timeRangeComboBox: PySide2.QtWidgets.QComboBox, clearButtonChannelA: PySide2.QtWidgets.QPushButton, clearButtonChannelB: PySide2.QtWidgets.QPushButton, clearButtonChannelC: PySide2.QtWidgets.QPushButton, clearButtonChannelD: PySide2.QtWidgets.QPushButton, saveDataButton: PySide2.QtWidgets.QPushButton, savePlotButton: PySide2.QtWidgets.QPushButton, countChannelAValue: PySide2.QtWidgets.QLabel, countChannelBValue: PySide2.QtWidgets.QLabel, countChannelCValue: PySide2.QtWidgets.QLabel, countChannelDValue: PySide2.QtWidgets.QLabel, countChannelAUncertainty: PySide2.QtWidgets.QLabel, countChannelBUncertainty: PySide2.QtWidgets.QLabel, countChannelCUncertainty: PySide2.QtWidgets.QLabel, countChannelDUncertainty: PySide2.QtWidgets.QLabel, tableCounts: PySide2.QtWidgets.QTableWidget, graphicsFrame: PySide2.QtWidgets.QFrame,*

Class responsible for managing the logic and graphical representation of the Count Estimation window.

This class handles user interactions and graphical updates related to photon/event count estimations across multiple channels (A, B, C, D). It coordinates the initialization of graphical elements, manages real-time measurements, and provides options for data saving, graphical visualization modes, and uncertainty tracking per channel. The main responsibilities include: - Initializing and updating graphical frames for each measurement channel. - Starting and stopping the count measurements. - Handling user interactions with checkboxes, radio buttons, and combo boxes. - Displaying and updating count values and their uncertainties. - Managing separate, merged, or detached graphics. - Saving count data and graphical plots. - Monitoring the device connection and measurement state.

Parameters

- `channelACheckBox` – Checkbox for enabling/disabling measurements on channel A (QCheckBox).
- `channelBCheckBox` – Checkbox for enabling/disabling measurements on channel B (QCheckBox).
- `channelCCheckBox` – Checkbox for enabling/disabling measurements on channel C (QCheckBox).
- `channelDCheckBox` – Checkbox for enabling/disabling measurements on channel D (QCheckBox).
- `startButton` – Button to start the count measurement (QPushButton).
- `stopButton` – Button to stop the count measurement (QPushButton).
- `mergeRadio` – Radio button to merge all channel plots into one graph (QRadioButton).
- `separateGraphics` – Radio button to display plots separately per channel (QRadioButton).
- `deatachedGraphics` – Radio button to display detached external graphics (QRadioButton).
- `timeRangeComboBox` – Combo box for selecting the time range for plot visualization (QComboBox).
- `clearButtonChannelA` – Button to clear the graph and data of channel A (QPushButton).
- `clearButtonChannelB` – Button to clear the graph and data of channel B (QPushButton).
- `clearButtonChannelC` – Button to clear the graph and data of channel C (QPushButton).
- `clearButtonChannelD` – Button to clear the graph and data of channel D (QPushButton).
- `saveDataButton` – Button to save count data for all channels (QPushButton).
- `savePlotButton` – Button to save the plot images (QPushButton).
- `countChannelAValue` – Label displaying current count value on channel A (QLabel).
- `countChannelBValue` – Label displaying current count value on channel B (QLabel).
- `countChannelCValue` – Label displaying current count value on channel C (QLabel).
- `countChannelDValue` – Label displaying current count value on channel D (QLabel).
- `countChannelAUncertainty` – Label showing uncertainty of channel A (QLabel).
- `countChannelBUncertainty` – Label showing uncertainty of channel B (QLabel).
- `countChannelCUncertainty` – Label showing uncertainty of channel C (QLabel).
- `countChannelDUncertainty` – Label showing uncertainty of channel D (QLabel).

- **tableCounts** – Table displaying count and uncertainty data per measurement cycle (QTableWidget).
- **graphicsFrame** – Frame container for holding the graphical layouts (QFrame).
- **channelAFrameLabel** – Frame label associated with channel A graphics (QFrame).
- **channelBFrameLabel** – Frame label associated with channel B graphics (QFrame).
- **channelCFrameLabel** – Frame label associated with channel C graphics (QFrame).
- **channelDFrameLabel** – Frame label associated with channel D graphics (QFrame).
- **statusLabel** – Label showing the system status (QLabel).
- **pointStatusLabel** – Label showing the status of the current measurement point (QLabel).
- **deatachedCheckBox** – Checkbox to toggle between embedded and detached graphics layout (QCheckBox).
- **detachedLabelCheckBox** – Checkbox to toggle detached labels in the plot layout (QCheckBox).
- **helpButton** – Button that displays a help/information dialog (QPushButton).
- **device** – The connected measurement device.
- **parent** – The parent window containing the UI (usually a QMainWindow).
- **timerConnection** – Timer responsible for checking device connection status (QTimer).

captureFinalDate(*date*)

Stores the final date received from a signal.

Parameters **date** – QDateTime - The end date of the measurement range.

Returns None

captureInitalDate(*date*)

Stores the initial date received from a signal.

Parameters **date** – QDateTime - The start date of the measurement range.

Returns None

captureMeasurement(*secondsTime*, *dateTime*, *channelAValue*, *channelAUncertainty*, *channelBValue*, *channelBUncertainty*, *channelCValue*, *channelCUncertainty*, *channelDValue*, *channelDUncertainty*)

Captures and processes a new set of measurement values emitted via signal, updating the UI and internal data structures.

This function is connected to a QSignal from the background worker thread and is responsible for handling a single measurement cycle. It validates the presence of data, provides real-time user feedback when no measurements are detected for specific channels, updates plots with valid values, populates the measurement table, and refreshes visual labels.

This function performs the following actions: - Checks for zero values in any channel and informs the user if one or more channels are not receiving valid measurements. - Updates the status bar and color indicator depending on whether all channels are acquiring data. - Appends valid count values and timestamps to the corresponding lists for each channel. - Updates the graphical curves for each channel if valid data is present. - Converts zero and -1 values into status messages like "Low Counts" or "Not Selected" for display purposes. - Adds the latest measurement row to both the main table (*tableCounts*) and its clone (*cloneTable*). - Updates the live count and uncertainty labels for any actively selected channel. - Refreshes the visible X-axis range of the graphs using *updateGraphic()*.

Parameters

- **secondsTime** – Timestamp in seconds since measurement start (float).
- **dateTime** – Human-readable datetime string for the measurement (str).
- **channelAValue** – Count value for channel A (float or int).
- **channelAUncertainty** – Uncertainty value for channel A (float).
- **channelBValue** – Count value for channel B (float or int).
- **channelBUncertainty** – Uncertainty value for channel B (float).
- **channelCValue** – Count value for channel C (float or int).
- **channelCUncertainty** – Uncertainty value for channel C (float).
- **channelDValue** – Count value for channel D (float or int).
- **channelDUncertainty** – Uncertainty value for channel D (float).

Returns None

changeStatusColor(*color*)

Changes the color of the point in the status bar. Each color is assigned a specific numeric value.

Parameters **color** – The numeric value corresponding to the desired color (int). - 0: Gray - 1: Green - 2: Yellow - 3: Orange

Returns None

changeStatusLabel(*textValue*)

Updates the status label with a new message.

This function sets the text of the *statusLabel* widget to the given string. It is used throughout the application to inform the user of the current measurement status, errors, warnings, or progress.

Parameters **textValue** – str - The message to display in the status label.

Returns None

checkBoxListenerChannels()

Updates the visibility of graphical elements and table columns based on the state of the channel checkboxes.

This function performs the following actions: - Checks the state of each channel's checkbox (A, B, C, D). - Shows or hides the corresponding frame label for each channel depending on whether the checkbox is checked. - Calls *hideColumns()* to hide the table columns associated with unchecked channels. - Calls *updateGraphicsLayout()* to refresh the displayed plots based on the selected channels.

Returns None

clearChannelA()

Clears all stored measurement data for channel A and updates the corresponding plot.

This function is called when the user clicks the Clear button for channel A. It resets the timestamp lists and count values associated with channel A, and refreshes the plot to reflect the cleared state.

This function performs the following actions: - Empties the list of raw timestamps and formatted date timestamps for channel A. - Clears the list of count values associated with channel A. - Updates the channel A curve to remove all plotted data.

Returns None

clearChannelB()

Clears all stored measurement data for channel B and updates the corresponding plot.

This function is called when the user clicks the Clear button for channel B. It resets the timestamp lists and count values associated with channel B, and refreshes the plot to reflect the cleared state.

This function performs the following actions: - Empties the list of raw timestamps and formatted date timestamps for channel B. - Clears the list of count values associated with channel B. - Updates the channel B curve to remove all plotted data.

Returns None

`clearChannelC()`

Clears all stored measurement data for channel C and updates the corresponding plot.

This function is called when the user clicks the Clear button for channel C. It resets the timestamp lists and count values associated with channel C, and refreshes the plot to reflect the cleared state.

This function performs the following actions: - Empties the list of raw timestamps and formatted date timestamps for channel C. - Clears the list of count values associated with channel C. - Updates the channel C curve to remove all plotted data.

Returns None

`clearChannelD()`

Clears all stored measurement data for channel D and updates the corresponding plot.

This function is called when the user clicks the Clear button for channel D. It resets the timestamp lists and count values associated with channel D, and refreshes the plot to reflect the cleared state.

This function performs the following actions: - Empties the list of raw timestamps and formatted date timestamps for channel D. - Clears the list of count values associated with channel D. - Updates the channel D curve to remove all plotted data.

Returns None

`closeDialogChannels(channel)`

Handles cleanup and UI updates when a detached graphics dialog is closed.

This function is triggered when a user manually closes a detached dialog window corresponding to a specific channel. It clears the internal reference to the dialog and unchecks the associated channel checkbox to reflect that the channel is no longer active in the detached view.

This function performs the following actions: - Sets the corresponding *dialogXCreated* variable to *None*, where X is the channel (A–D). - Unchecks the checkbox for the specified channel to visually indicate it is no longer selected.

Parameters `channel` – The identifier of the channel whose dialog was closed ('A', 'B', 'C', or 'D').

Returns None

`closeLabelDialog()`

Resets the dialog state and unchecks the measurement label checkbox if the dialog was closed while still marked as active.

Returns None

`closeTableDialog()`

Handles cleanup when the detached table dialog is closed.

This function is triggered when the detached table dialog (*dialogTableOpen*) is closed by the user. It performs the following actions to maintain consistency in the UI state:

- Sets the internal dialog reference *dialogTableOpen* to *None*.
- Unchecks the *deatachedCheckBox* to reflect that the detached view is no longer active.

This ensures that the application's interface remains synchronized and avoids inconsistencies when reopening or reattaching the measurement table view.

Returns None

`connectedDevice`(*device*)

Handles the UI and internal state updates when the device is successfully connected.

This function performs the following actions: - Enables the disconnect button and disables the connect button in the main window. - Updates the internal sentinel to indicate that the device is connected and measurements are allowed. - Stores the reference to the connected device. - Enables the start button to allow measurements to begin. - (Optional) Starts a timer for updating measurement status (currently commented out).

Parameters `device` – The connected device instance.

Returns None

`constructAllGraphics`()

Constructs and initializes the unified graphical layout for all measurement channels.

This function calls *factoryGraphsAllChannels* to generate a single plot widget that contains curves for channels A, B, C, and D. Each curve is customized by color and visibility based on whether its corresponding checkbox is enabled. The resulting components are stored as instance variables for centralized access and updates.

This function performs the following actions: - Calls *factoryGraphsAllChannels* to create the shared plot and individual channel curves. - Stores the plot widget, plot item, and all channel-specific curves as instance variables.

Returns None

`constructGraphicA`()

Constructs and initializes the graphical components for channel A.

This function uses the *factoryGraphChannels* method to generate the graphical layout for channel A, including the main plot widget, the plot area, and the curve used to display count data. The resulting components are stored as instance variables for later access and updates during measurement.

This function performs the following actions: - Calls *factoryGraphChannels* with channel 'A' to generate the graph. - Stores the returned plot widget, plot item, and curve for channel A.

Returns None

`constructGraphicB`()

Constructs and initializes the graphical components for channel B.

This function uses the *factoryGraphChannels* method to generate the graphical layout for channel B, including the main plot widget, the plot area, and the curve used to display count data. The resulting components are stored as instance variables for later access and updates during measurement.

This function performs the following actions: - Calls *factoryGraphChannels* with channel 'B' to generate the graph. - Stores the returned plot widget, plot item, and curve for channel B.

Returns None

`constructGraphicC`()

Constructs and initializes the graphical components for channel C.

This function uses the *factoryGraphChannels* method to generate the graphical layout for channel C, including the main plot widget, the plot area, and the curve used to display count data. The resulting components are stored as instance variables for later access and updates during measurement.

This function performs the following actions: - Calls *factoryGraphChannels* with channel 'C' to generate the graph. - Stores the returned plot widget, plot item, and curve for channel C.

Returns None

`constructGraphicD()`

Constructs and initializes the graphical components for channel D.

This function uses the *factoryGraphChannels* method to generate the graphical layout for channel D, including the main plot widget, the plot area, and the curve used to display count data. The resulting components are stored as instance variables for later access and updates during measurement.

This function performs the following actions: - Calls *factoryGraphChannels* with channel 'D' to generate the graph. - Stores the returned plot widget, plot item, and curve for channel D.

Returns None

`createCloneTable()`

Creates and configures a cloned table instance for storing a copy of the measurement data.

This function initializes a separate *QTableWidget* that mirrors the structure of the main count table. It includes predefined column headers for the date and channels A, B, C, and D. The cloned table is set to stretch its columns to fit the available space and is made read-only to prevent user edits.

This function performs the following actions: - Creates a new *QTableWidget* instance with 5 columns. - Sets the horizontal headers to ['Date', 'A', 'B', 'C', 'D']. - Enables automatic stretching of column widths. - Disables editing to preserve the integrity of the data.

Returns None

`createDialogFactory(`*channel*`)`

Creates and returns a non-modal dialog window for displaying detached graphics for the specified channel.

This factory function is used when the user selects the "detached graphics" mode. It generates a dedicated *QDialog* for a given channel (A, B, C, or D), configures its appearance and behavior, and positions it to avoid overlap with other dialogs. The dialog is linked to a callback that ensures proper cleanup when it is closed.

This function performs the following actions: - Instantiates a *QDialog* as a child of the main window. - Sets the dialog title and default size. - Connects the *finished* signal to a cleanup method for the channel. - Assigns a fixed screen position for each channel to avoid visual overlap.

Parameters `channel` – A string indicating the target channel ('A', 'B', 'C', or 'D').

Returns A configured, ready-to-display *QDialog* for the given channel.

`createRow(`*leftLabel*, *rightLabel*`)`

Creates a horizontal row frame containing two labels.

This function constructs a *QFrame* that represents a single row in a dialog or panel. It arranges two given *QLabel* widgets — one aligned to the left and the other to the right — using a *QHBoxLayout*. The labels are styled with a consistent font size and the vertical margins are removed to minimize vertical spacing between rows.

This method is used to visually structure measurement data (such as estimated counts and uncertainties) in a clean, compact format.

Parameters

- `leftLabel` – QLabel - The label to place on the left side of the row.
- `rightLabel` – QLabel - The label to place on the right side of the row.

Returns QFrame - A frame containing the two labels arranged horizontally.

`deatachedTable()`

Manages the visibility of the estimated counts table in a detached dialog window.

This function is triggered when the *deatachedCheckBox* is toggled. If the checkbox is checked, it opens a new non-modal *QDialog* containing the cloned measurement table (*cloneTable*). If the checkbox is unchecked, it closes the dialog and resets the internal reference to *None*.

This function performs the following actions: - If the detached table mode is enabled:

- Creates and configures a new *QDialog* with the cloned table.
- Displays the dialog to the user.
- Connects the dialog's close event to a cleanup function.
- **If the detached table mode is disabled:**
  - Closes the existing dialog if it is open.
  - Clears the internal dialog reference.

Returns None

`detachedLabels()`

Displays or hides a dialog showing current measurement values for selected channels.

When the 'Detached current measurement' checkbox is checked, this function creates and displays a non-modal QDialog containing the current estimated count rates and their uncertainties for each selected channel (A–D). Each visible row in the dialog corresponds to a channel, displaying the estimated counts (in cps) and the associated uncertainty. If a channel is not selected, its row remains hidden.

The dialog layout is organized using a sunken panel-style QFrame with a vertical layout. A bold header row indicates the meaning of each column. This interface allows users to quickly inspect the most recent count values without modifying the ongoing measurement.

If the checkbox is unchecked, any open measurement dialog is closed automatically.

Returns None

`disconnectedDevice()`

Handles the UI and internal state updates when the device is disconnected.

This function performs the following actions: - Disables the disconnect button and enables the connect button in the main window. - Disables the start button to prevent new measurements from being initiated. - (Optional) May start a timer for monitoring status or reconnection attempts (currently commented out).

Returns None

`eliminateCheckBoxChannels(channelList)`

Handles the removal of channels with insufficient measurement signals and prompts the user to continue or stop.

This function is triggered when one or more selected channels fail to meet the minimum pulse requirements for reliable count estimation (e.g., fewer than 500 pulses per second). It presents a dialog informing the user about the affected channels and asks whether to continue the measurement with the remaining valid channels or to stop entirely.

This function performs the following actions: - Builds a human-readable list of the channels that failed to provide sufficient data. - Displays a warning dialog with the option to continue or cancel the measurement. - If the user chooses to continue:

- Disables the affected channels in the device.

- Unchecks and disables the associated checkboxes.
- Disables the "Clear" button for each affected channel.
- Marks the *measurementChannelX* sentinel as *False* for each affected channel.
- **If the user chooses not to continue:**
  - Stops the measurement thread.
- Signals the worker thread to proceed based on the user's decision via *continueEvent*.

Parameters `channelList` – A list of channel identifiers (e.g., ['A', 'C']) that failed to meet the minimum measurement conditions.

Returns None

`enableButtons()`

Enables or disables the clear buttons for each channel based on the current selection state.

This function checks which channels (A–D) have been marked as selected for measurement and enables their corresponding "Clear" buttons. If a channel is not selected, its clear button remains disabled. Additionally, it disables the disconnect button to prevent accidental disconnection during an active measurement setup.

This function performs the following actions: - Enables the "Clear" button for each selected channel. - Disables the "Clear" button for any unselected channel. - Disables the disconnect button in the main window.

Returns None

`factoryGraphChannels(channel)`

Creates and returns a configured plot widget for the specified measurement channel.

This factory function generates a graphical layout for visualizing count data from one of the four available channels (A, B, C, or D). The plot is customized with a unique color per channel, grid display, axis labeling, and a legend. It returns the PyQtGraph components necessary for real-time plotting.

This function performs the following actions: - Selects a predefined color for the specified channel. - Initializes a *GraphicsLayoutWidget* with a white background. - Adds a plot to the widget with grid lines, axis labels, and legend. - Creates a stylized curve object for plotting count data using markers and colored lines. - Returns the components needed to integrate the plot into the GUI and update it dynamically.

Parameters `channel` – A single-character string indicating the channel ('A', 'B', 'C', or 'D').

Returns A tuple containing: - *GraphicsLayoutWidget*: the plot container. - *PlotItem*: the configured plot. - *PlotDataItem*: the curve used for plotting the counts.

`factoryGraphsAllChannels()`

Creates and returns a single plot widget containing curves for all selected measurement channels.

This factory function generates a unified graphical layout that displays the estimated counts for channels A, B, C, and D in a single plot. Each channel is represented with a distinct color and labeled curve, depending on whether its checkbox is enabled. This approach consolidates the visualization into one graph rather than creating separate ones per channel.

This function performs the following actions: - Defines a unique color for each channel (A, B, C, D). - Initializes a *GraphicsLayoutWidget* with a white background and grid lines. - Configures a single plot with appropriate axis labels and a legend. - Creates curve objects for each channel using stylized markers and pens. - Only assigns a name to a curve if its corresponding channel checkbox is checked. - Returns the complete widget and all channel curves for display and real-time updates.

Returns A tuple containing: - *GraphicsLayoutWidget*: the shared plot container for all channels. - *PlotItem*: the unified plot area. - *PlotDataItem*: curve for channel A. - *PlotDataItem*: curve for channel B. - *PlotDataItem*: curve for channel C. - *PlotDataItem*: curve for channel D.

`finishedThread()`

Handles the post-execution cleanup and UI updates when the measurement thread finishes.

This function is automatically triggered when the background measurement thread completes. It resets internal sentinels, updates the UI to reflect the end of the measurement, enables or disables appropriate buttons depending on whether data was captured, and invokes the standard stop measurement routine.

This function performs the following actions: - Resets all *selectChannelX* sentinels (A–D) to *False*. - Enables the Start button and disables the Stop button. - If any channel has collected data, enables the save buttons for data and plots. - Disables all "Clear" buttons for individual channels. - Updates the status label and color to indicate that no measurement is running. - Calls *stopMeasure()* to finalize the stop process and restore the interface to its initial state.

Returns None

`getChannelsMeasure()`

Determines which measurement channels are selected and updates internal sentinels accordingly.

This function evaluates the state of each channel's checkbox (A, B, C, D) to establish which channels will participate in the measurement. It sets the corresponding selection and measurement sentinels to *True* or *False* based on the checkboxes. Additionally, it disables the checkbox for any unselected channel to prevent user changes during the active measurement session.

This function performs the following actions: - Assumes all channels are selected by default. - Checks the state of each channel checkbox and updates:

- *selectChannelX*: sentinel to include channel X in the measurement loop.
- *measurementChannelX*: sentinel to track measurement activity per channel.
- Disables the checkbox for any channel that was not selected to lock in the configuration.

Returns None

`helpButtonDialog()`

Displays an informational dialog with instructions for using the Count Estimation window.

This function shows a modal QMessageBox with an information icon to guide the user on how to correctly connect the signal sources for pulse estimation. It clarifies that a periodic signal should be connected to the Start input, and the sources to be measured should be connected to the Stop channels.

Returns None

`hideColumns()`

Updates the visibility of the table columns based on the selected measurement channels.

This function ensures that only the columns corresponding to active channels (A–D) are visible in both the main and cloned data tables. It first shows all channel columns, then hides those whose associated checkboxes are unchecked.

This function performs the following actions: - Ensures all channel columns (1–4) are visible initially in both *tableCounts* and *cloneTable*. - Checks the state of each channel's checkbox. - Hides the column in both tables for any channel that is not currently selected.

Returns None

`lostConnection()`

Displays a critical error message indicating that the device connection has been lost.

This function sets the sentinel flag *disconnectedMeasurement* to True and shows a modal QMessageBox with a critical icon to inform the user that the connection to the device has been lost. The dialog must be acknowledged before continuing.

Returns None

`noChannelsSelected()`

Displays a warning dialog indicating that no channels have been selected for measurement.

This function is called when the user attempts to start a measurement without selecting any channels. It generates a modal warning dialog to inform the user that at least one channel must be selected to proceed.

This function performs the following actions: - Opens a *QMessageBox* warning with a title and informative message. - Prevents the measurement process from starting until the issue is resolved.

Returns None

`noMeasurementsFounded()`

Displays a warning dialog and disables channel interactions when no measurements are detected.

This function is triggered when the system fails to obtain valid measurements from any of the selected channels. It informs the user of the issue, disables the "Clear" buttons for all channels, and resets the internal sentinels that indicate active measurement states.

This function performs the following actions: - Shows a *QMessageBox* warning explaining that no measurements were found and outlines the minimum requirements for detection (e.g., 500 pulses per second). - Disables the "Clear" buttons for channels A–D to prevent further user interaction. - Resets *measurementChannelX* sentinels (A–D) to *False*, marking all channels as inactive.

Returns None

`resetSaveSentinels()`

`resetSentinels()`

Resets internal channel selection sentinels and re-enables all channel checkboxes.

This function is typically called after a measurement is stopped. It clears the selection flags (*selectChannelX*) for all channels (A–D) and re-enables their corresponding checkboxes in the UI, allowing the user to configure new measurement selections.

This function performs the following actions: - Sets all *selectChannelX* sentinels to *False*, indicating no active selections. - Enables the checkboxes for channels A, B, C, and D to restore full configuration flexibility.

Returns None

`resetValues()`

Resets all measurement-related data, graphics, tables, and save-state sentinels to their initial state.

This function is typically called before starting a new measurement session. It ensures that all previous data and UI elements are cleared to avoid inconsistencies or overlapping results. It resets internal storage lists, clears the curves on the plots, empties the result tables, and resets file save status indicators.

This function performs the following actions: - Clears all count values and timestamps for channels A–D. - Clears the corresponding date-formatted timestamp lists. - Resets the row count of the main and cloned data tables. - Clears all data from the plotted curves for each channel. - Resets internal sentinels that track whether the data has been saved in TXT, CSV, or DAT format.

Returns None

`returnSettings()`

Restores the previously saved configuration settings to the Tempico device.

This function is typically called after a measurement session ends. It applies the saved settings—previously captured via *saveSettings()*—back to the Tempico device for all channels (A–D). This includes restoring the number of runs, number of stops, mode, averaging cycles, and stop mask values. If any of the required variables were not initialized (e.g., due to a missing *saveSettings()* call), the function catches the *NameError* and prints the exception.

This function performs the following actions: - Restores the number of acquisition runs for the entire device. - Restores the number of averaging cycles for each channel. - Restores the number of stop events for each channel. - Restores the operation mode for each channel. - Restores the stop mask configuration for each channel. - Catches and prints *NameError* if any setting is undefined.

Returns None

`rightAlignedItem`(*value*)

Creates a QTableWidgetItem with right-aligned and vertically centered text.

This function takes a value, converts it to a string, wraps it in a QTableWidgetItem, and sets the text alignment to align the content to the right and center it vertically. It is used to ensure consistent formatting of numerical or textual data in table cells.

Parameters `value` (*Any*) – The value to be displayed in the table cell.

Returns A QTableWidgetItem with the specified alignment.

Return type QTableWidgetItem

`saveData()`

Saves the current data according to the selected format specified in the dialog box.

The function starts by retrieving the default histogram name and the current date, which is formatted to create a unique filename. It then initializes lists to hold filenames, data, settings, and column names for the saved files.

A dialog box is displayed for the user to select the desired file format (txt, csv, or dat) for saving the data. Once the user accepts the selection, the function checks if the selected format has already been saved before.

If the selected format has not been saved, it collects data from the various channels (A, B, C, D) if their corresponding sentinel variables indicate they should be saved. It retrieves the average cycles, mode, number of stops, stop edge, and stop mask for each channel to include in the settings. The function then attempts to save the collected data in the specified format.

If the save operation is successful, a message box is displayed, confirming the successful save and showing the folder path and file names. If an error occurs during the save process, an error message box is shown.

If the selected format has already been saved, a message box is displayed with the previous save information.

Returns None

`savePlots()`

Saves the current plots in the selected image format.

This method opens a dialog for the user to select an image format (PNG, TIFF, or JPG) and saves the plots for channels A, B, C, and D if their respective flags are set to True. The plots are saved with a timestamp in the specified format in the default folder path.

Returns None

`saveSettings()`

Saves the current configuration settings of the Tempico device for all channels.

This function queries the device and stores key configuration parameters for each channel (A–D), such as the number of stops, acquisition mode, averaging cycles, and stop mask. These values are cached in instance

variables so they can be restored later (e.g., after a measurement session). If the device is unavailable or any read operation fails, the function safely ignores the error.

This function performs the following actions: - Retrieves and stores the global number of acquisition runs. - Retrieves and stores the number of stops per channel. - Retrieves and stores the operation mode per channel. - Retrieves and stores the number of averaging cycles per channel. - Retrieves and stores the stop mask per channel. - Ignores exceptions silently if the device is not ready.

Returns None

`startMeasure()`

Starts the count measurement process by configuring the UI, preparing internal states, and launching the background worker thread.

This function is triggered when the user clicks the Start button. It verifies that at least one measurement channel (A–D) is selected. If so, it proceeds to initialize the measurement by disabling conflicting buttons and tabs, saving current settings, resetting values, and setting up the worker thread responsible for data acquisition. If no channels are selected, it displays a dialog warning the user.

This function performs the following actions: - Verifies that at least one channel checkbox is active. - Disables the Start button and enables the Stop button. - Disables data saving buttons and tab navigation to prevent user interference. - Saves current settings and stops the connection monitoring timer. - Resets measurement values and identifies the selected channels. - Updates the UI status bar to indicate active measurement. - Creates and configures a *WorkerThreadCountsEstimated* thread to perform background data acquisition. - Connects various thread signals to corresponding update methods in the class. - Starts the measurement thread. - If no channels are selected, displays a warning dialog requiring the user to activate at least one channel.

Returns None

`startTimerConnection()`

Starts the connection monitoring timer to periodically check the device status.

This function activates the *timerConnection*, which verifies the connection with the Tempico device every 500 milliseconds. It is typically called after a measurement has ended or when the device is reconnected, ensuring continuous monitoring of the device's availability.

Returns None

`stopMeasure()`

Stops the ongoing measurement process and restores the UI and device to their initial configuration.

This function is triggered when the user clicks the Stop button. It halts the background measurement thread, resets relevant internal flags (sentinels), re-enables UI tabs and controls, and restores the initial settings of the Tempico device if the disconnection did not occur. If the device was disconnected during measurement, it triggers the disconnection handling routine.

This function performs the following actions: - Re-enables the disabled UI tabs to allow user interaction post-measurement. - Calls *resetSentinels()* to clear internal flags related to active measurement channels. - Disables the Stop button and signals the worker thread to stop. - If the device is still connected:

- Restarts the connection-monitoring timer.
- Re-enables the disconnect button.
- **If the device was disconnected during measurement:**
  - Triggers the disconnection handling logic in the main window.
- Updates the main window status to indicate that no measurement is currently running.
- Calls *returnSettings()* to restore device parameters to their original configuration.

Returns None

`stopTimerConnection()`

Stops the connection monitoring timer during an active measurement.

This function is called when a measurement begins to prevent interference from the periodic connection checks. It halts the *timerConnection*, which normally runs in the background to monitor the status of the Tempico device.

Returns None

`updateGraphic()`

Updates the visible X-axis range of all channel plots based on the selected time window.

This function is triggered when the user changes the time range using the *timeRangeComboBox*. It determines the most recent timestamp in channel A and adjusts the X-axis range for all plots (A–D) accordingly, displaying only the data within the selected time interval.

This function performs the following actions: - Verifies that channel A has data; if not, it exits early. - Attempts to extract the number of seconds from the selected combo box option. - Calculates the minimum and maximum X-axis values based on the most recent timestamp. - Updates the X-axis range of all plots (A, B, C, and D) to reflect the selected time window.

Returns None

`updateGraphicsLayout()`

Updates the graphical layout based on the selected visualization mode and active channel checkboxes.

This function dynamically reconfigures how the channel plots are displayed, depending on the selected radio button mode: separate graphics, merged graph, or detached dialogs. It ensures that the interface reflects the current user selections by showing or hiding the corresponding curves, windows, or layouts.

This function performs the following actions: - If **separate graphics** mode is selected:

- Closes any open detached dialogs.
- Clears the current layout inside the *graphicsFrame*.
- Creates individual graphs for each selected channel (A–D).
- **Arranges them responsively:**
  - 1 channel → full frame.
  - 2 channels → one above the other.
  - 3 channels → two top, one bottom.
  - 4 channels → 2×2 grid layout.
- **If merge graphics mode is selected:**
  - Closes detached dialogs if any exist.
  - Clears the layout in *graphicsFrame*.
  - Creates a single graph containing all curves.
  - Displays or hides each curve depending on its checkbox status.
- **If detached graphics mode is selected:**
  - Clears the main *graphicsFrame* layout.
  - For each selected channel (A–D), opens a dedicated dialog window.

- Loads previous data into the newly created plots.
- Closes dialog windows for unchecked channels.

Returns None

`updateLabels`(*channel*, *value*, *uncertainty*)

Updates the UI labels displaying the latest measurement value and its uncertainty for the specified channel.

This function formats and sets the value and uncertainty labels for a given channel (A–D). If the value is a string (e.g., "Low Counts" or "Not Selected"), it is displayed as-is. If it is a numeric value, the function rounds it for clearer presentation before updating the corresponding labels.

This function performs the following actions: - Determines if the value is a string or numeric. - Rounds numeric values (value to 2 decimals, uncertainty to 5 decimals). - Updates the *QLabel* elements associated with the selected channel to reflect the latest measurement results.

Parameters

- `channel` – The target channel identifier ('A', 'B', 'C', or 'D').
- `value` – The most recent measured value (float, int, or str).
- `uncertainty` – The associated uncertainty of the measurement (float or str).

Returns None

CHAPTER

FIVE

TIMESTAMPLOGIC MODULE

class Logic.TimeStampLogic.**TimeStampLogic**(*enableCheckBoxA: PySide2.QtWidgets.QCheckBox*, *enableCheckBoxB: PySide2.QtWidgets.QCheckBox*, *enableCheckBoxC: PySide2.QtWidgets.QCheckBox*, *enableCheckBoxD: PySide2.QtWidgets.QCheckBox*, *startNormalButton: PySide2.QtWidgets.QPushButton*, *pauseNormalButton: PySide2.QtWidgets.QPushButton*, *stopNormalButton: PySide2.QtWidgets.QPushButton*, *startScheduleButton: PySide2.QtWidgets.QPushButton*, *pauseScheduleButton: PySide2.QtWidgets.QPushButton*, *stopScheduleButton: PySide2.QtWidgets.QPushButton*, *startLimitedButton: PySide2.QtWidgets.QPushButton*, *pauseLimitedButton: PySide2.QtWidgets.QPushButton*, *stopLimitedButton: PySide2.QtWidgets.QPushButton*, *startDate: PySide2.QtWidgets.QDateEdit*, *startTime: PySide2.QtWidgets.QTimeEdit*, *finishDate: PySide2.QtWidgets.QDateEdit*, *finishTime: PySide2.QtWidgets.QTimeEdit*, *numberMeasurementsSpinBox: PySide2.QtWidgets.QSpinBox*, *showTableCheckBox: PySide2.QtWidgets.QCheckBox*, *measurementLabelA: PySide2.QtWidgets.QLabel*, *measurementLabelB: PySide2.QtWidgets.QLabel*, *measurementLabelC: PySide2.QtWidgets.QLabel*, *measurementLabelD: PySide2.QtWidgets.QLabel*, *valueMeasurementA: PySide2.QtWidgets.QLabel*, *valueMeasurementB: PySide2.QtWidgets.QLabel*, *valueMeasurementC: PySide2.QtWidgets.QLabel*, *valueMeasurementD: PySide2.QtWidgets.QLabel*, *valueTotalMeasurement: PySide2.QtWidgets.QLabel*, *tableTimeStamp: PySide2.QtWidgets.QTableWidget*, *statusLabel: PySide2.QtWidgets.QLabel*, *colorLabel: PySide2.QtWidgets.QLabel*, *saveDataComplete: PySide2.QtWidgets.QCheckBox*, *tabNormalMeasurement: PySide2.QtWidgets.QWidget*, *tabScheduleMeasurement: PySide2.QtWidgets.QWidget*, *tabLimitedMeasurement: PySide2.QtWidgets.QWidget*, *saveDataButton: PySide2.QtWidgets.QPushButton*, *tabs: PySide2.QtWidgets.QTabWidget*, *autoSaveComboBox: PySide2.QtWidgets.QComboBox*, *helpSaveButton: PySide2.QtWidgets.QPushButton*, *parent*, *device: pyTempico.core.TempicoDevice*, *timerConnection: PySide2.QtCore.QTimer*)

Bases: object

Class responsible for managing the logic and interface of the TimeStamp measurement tab.

This class handles long-duration timestamp measurements across different operational modes, including: - **Normal** mode: Starts and stops measurements manually. - **Scheduled** mode: Automatically starts/stops measurements based on configured start and end date-times. - **Limited** mode: Runs measurements until a predefined number of events is recorded.

It provides real-time updates of the measurement status, displays results in a table, and supports manual and automatic data saving in multiple formats. It also offers visual feedback, per-channel control, and user configuration for saving behaviors.

The main responsibilities include: - Managing three measurement modes: normal, scheduled, and limited. - Displaying measurement data in a table, including per-channel and total counts. - Handling delayed start and stop based on date and time settings. - Enabling saving of data manually or automatically in *.txt*, *.csv*, and *.dat* formats. - Tracking system and measurement status through visual and textual indicators. - Controlling visibility and behavior of measurement labels and checkboxes. - Providing help dialogs and UI feedback during configuration.

Parameters

- `enableCheckBoxA` – Checkbox to enable/disable channel A (QCheckBox).
- `enableCheckBoxB` – Checkbox to enable/disable channel B (QCheckBox).
- `enableCheckBoxC` – Checkbox to enable/disable channel C (QCheckBox).
- `enableCheckBoxD` – Checkbox to enable/disable channel D (QCheckBox).
- `startNormalButton` – Button to start normal measurement mode (QPushButton).
- `pauseNormalButton` – Button to pause normal measurement mode (QPushButton).
- `stopNormalButton` – Button to stop normal measurement mode (QPushButton).
- `startScheduleButton` – Button to start scheduled measurement mode (QPushButton).
- `pauseScheduleButton` – Button to pause scheduled measurement mode (QPushButton).
- `stopScheduleButton` – Button to stop scheduled measurement mode (QPushButton).
- `startLimitedButton` – Button to start limited measurement mode (QPushButton).
- `pauseLimitedButton` – Button to pause limited measurement mode (QPushButton).
- `stopLimitedButton` – Button to stop limited measurement mode (QPushButton).
- `startDate` – Date selector for scheduled start date (QDateEdit).
- `startTime` – Time selector for scheduled start time (QTimeEdit).
- `finishDate` – Date selector for scheduled end date (QDateEdit).
- `finishTime` – Time selector for scheduled end time (QTimeEdit).
- `numberMeasurementsSpinBox` – Spin box to configure number of measurements in limited mode (QSpinBox).
- `showTableCheckBox` – Checkbox to show/hide the measurement table (QCheckBox).
- `measurementLabelA` – Label title for channel A measurement (QLabel).
- `measurementLabelB` – Label title for channel B measurement (QLabel).
- `measurementLabelC` – Label title for channel C measurement (QLabel).
- `measurementLabelD` – Label title for channel D measurement (QLabel).
- `valueMeasurementA` – Label showing count value for channel A (QLabel).

- **valueMeasurementB** – Label showing count value for channel B (QLabel).
- **valueMeasurementC** – Label showing count value for channel C (QLabel).
- **valueMeasurementD** – Label showing count value for channel D (QLabel).
- **valueTotalMeasurement** – Label showing total count value (QLabel).
- **tableTimeStamp** – Table displaying recorded timestamp data (QTableWidget).
- **statusLabel** – Label showing current system status (QLabel).
- **colorLabel** – Color-coded label to indicate measurement state (QLabel).
- **saveDataComplete** – Checkbox to toggle full data saving after measurement (QCheckBox).
- **tabNormalMeasurement** – Tab widget for normal measurement UI (QWidget).
- **tabScheduleMeasurement** – Tab widget for scheduled measurement UI (QWidget).
- **tabLimitedMeasurement** – Tab widget for limited measurement UI (QWidget).
- **saveDataButton** – Button to manually save timestamp data (QPushButton).
- **tabs** – Main tab widget that hosts all measurement tabs (QTabWidget).
- **autoSaveComboBox** – Combo box to select preferred format for automatic saving (QComboBox).
- **helpSaveButton** – Button to show help dialog for save options (QPushButton).
- **parent** – The parent UI element containing this tab (usually QMainWindow).
- **device** – The connected TempicoDevice used for timestamp acquisition.
- **timerConnection** – Timer object for device connection polling (QTimer).

autoSaveAction()

Performs periodic autosave during a measurement.

This function is triggered automatically at regular intervals while a measurement is running. It saves the current batch of measurement data to a file using the predefined autosave format, then clears the saved data from memory to free resources.

Returns None

autoSaveActionRoute()

Performs the final autosave when a measurement ends.

This function is executed as the last step of a measurement when either the user stops it manually or the scheduled time finishes. It saves all remaining measurement data to a file using the selected autosave format (*txt*, *csv*, or *dat*), clears the in-memory data lists, and updates the corresponding flags indicating that the data has been saved.

Returns None

autoSaveSettings(*currentTab*)

Configures and starts the autosave timer based on the user's interval selection.

If not in the "Scheduled" tab, prompts the user to select the save format.

beginMeasurementTime()

Starts the scheduled timestamp measurement when the countdown timer reaches zero.

This function is automatically called after the waiting period in a scheduled measurement. It performs the following actions: - Clears previous data and resets internal sentinels. - Sets up pre-measurement values and stops the device connection timer. - Applies auto-save settings and generates the file name if needed.

- Initializes the measurement worker thread in "scheduled" mode. - Starts a timer to monitor when the measurement should stop automatically. - Updates the UI to reflect that the measurement is now running.

Returns None

captureMeasurement(*valuesA*, *valuesB*, *valuesC*, *valuesD*, *valuesStart*, *totalMeasurementsChannelA*, *totalMeasurementsChannelB*, *totalMeasurementsChannelC*, *totalMeasurementsChannelD*, *totalMeasurements*)

Inserts new measurement data into the table, updates the internal data structures, and refreshes channel/total measurement labels.

Called when the worker thread emits new measurement values for each channel.

- Adds rows for each measurement (A, B, C, D, and Start events) at the top of the table.
- Stores timestamps, stop values, and channel IDs in internal lists.
- Limits table size to 50,000 rows.
- Updates labels showing total measurements per channel and overall.

changeLabelToUpdate(*textValue*)

Updates the status label or measurement status value depending on the reordering state.

If not currently reordering, updates *statusValueMeasurements*. If reordering, updates the *statusLabel* text directly.

Parameters **textValue** (*str*) – The text to display.

Returns None

changeStatusColor(*color*)

Changes the color of the point in the status bar. Each color is assigned a specific numeric value.

Parameters **color** – The numeric value corresponding to the desired color (int). - 0: Gray - 1: Green - 2: Yellow - 3: Orange

Returns None

changeStatusLabel(*textValue*)

Updates the status label text unless a save operation is in progress.

If a save is occurring, the label is set to "Saving Data" instead.

Parameters **textValue** (*str*) – The text to display if not currently saving.

Returns None

checkBoxSaveData()

Handles the behavior of the "Save Data" checkbox for enabling or disabling auto-save options. Includes a confirmation dialog when disabling auto-save.

checkBoxSaveDataAfterMeasurement()

Handles the behavior of the "Save Data" checkbox for enabling or disabling auto-save options. Includes a confirmation dialog when disabling auto-save.

clearData()

Clears all data and resets save-related sentinels from the previous measurement.

This includes: - Resetting autosave and manual save flags. - Clearing stored measurement data and timestamps. - Resetting the display table. - Clearing internal autosave buffer via *savefile*.

Returns None

connectedDevice(*device*)

Handles UI and internal state updates when the device is connected in the TimeStamp logic tab.

This function performs the following actions: - Enables the disconnect button and disables the connect button in the main window. - Updates the internal sentinel to indicate the device is now connected. - Stores the reference to the connected device instance. - Enables the start buttons for normal, scheduled, and limited measurement modes.

Parameters **device** – The connected device instance.

Returns None

countToBegin()

Updates the countdown status before the scheduled measurement begins.

This function is called periodically by a QTimer during the waiting period before a scheduled measurement starts. It performs the following actions: - Calculates the remaining time until the scheduled start. - If time is still remaining: - Updates the status label and color with a formatted countdown. - If the start time has been reached or passed: - Stops the timer. - Calls the function to begin the actual measurement (*beginMeasurementTime*).

Returns None

countToStop()

Updates the countdown status until the scheduled measurement ends.

This function is called periodically by a QTimer during an active scheduled measurement. It performs the following actions: - Calculates the remaining time until the configured end time. - If time is still remaining: - Updates the status label with the current measurement state and formatted time. - If the end time has been reached or passed: - Automatically stops the scheduled measurement by calling *stopScheduledMeasurement()*.

Returns None

dialogFormatStart(*currentTab*)

Displays a dialog for selecting the file format when autosave is enabled.

This function is called at the start of a measurement to let the user choose the file format (*txt*, *csv*, or *dat*) for automatic data saving. If the user accepts, the selection is stored and measurement continues. If the user cancels, the measurement is stopped according to the current tab (*Normal*, *Scheduled*, or *Limited*) and common post-measurement settings are restored.

Parameters **currentTab** (*str*) – The tab where the measurement was started ("Normal", "Scheduled", or "Limited").

dialogHelpSaveButton()

Opens a help dialog explaining how autosave works.

Returns None

dialogToShowSave()

Displays a confirmation dialog for a successful save operation.

This function shows an information dialog indicating that the files have been saved successfully. It includes the save folder path and the generated file name based on the measurement start date and selected format.

Returns None

disconnectedDevice()

Handles UI and internal state updates when the device is disconnected.

This function performs the following actions: - Resets the internal sentinel indicating whether data is currently being saved. - Updates the status label and color to reflect that no measurement is running. - Disables

the disconnect button and enables the connect button in the main window. - Disables the start buttons for normal, scheduled, and limited measurement modes.

Returns None

`fileNameAutoSave()`

Generates and stores the autosave file name.

This function builds the complete file path for the autosave operation based on the save folder, file prefix, measurement start date, and selected file format. The resulting path is stored in the *fileName* class attribute for later use.

Returns None

`finalMessageAfterSave()`

Displays a confirmation dialog indicating that the file was saved successfully.

This function shows an information message box with the final save message (*self.saveFinalText*), informing the user of the file's location and name.

Returns None

`finishedMeasurements()`

`finishedThread()`

Handles post-measurement actions after the normal measurement thread finishes.

If autosave is enabled, prompts the user to save. Then resets saving state and stops the normal measurement.

`finishedThreadLimited()`

`finishedThreadSchedule()`

`formatWaitingTime(secondsTotal)`

Formats a time duration (in seconds) into a human-readable string for countdown display.

This function is used to show the remaining time before a scheduled measurement starts or ends. The format adapts depending on the total duration: - If more than one day: *"Xd HH:MM:SS"* - If more than one hour: *"H:MM:SS"* - If less than one hour: *"M:SS"*

Parameters `secondsTotal` – Total time in seconds to format.

Returns A formatted string representing the remaining time.

Return type str

`getAproxTotalData(availableRamMb)`

`getAvailableRam()`

`notSelectedFormatLimitedStop()`

Cancels the limited measurement setup when no save format is selected while auto-save is enabled.

This function is triggered when the user initiates a limited measurement with auto-save enabled but closes the format selection dialog without choosing a file format.

It performs the following actions: - Aborts the measurement setup and resets saving state. - Updates the status label and color to indicate no measurement is running. - Re-enables the start button and disables pause/stop controls. - Re-enables the measurement count spinbox. - Restores tab access for the local interface and main window. - Resumes the device connection timer.

Returns None

`notSelectedFormatNormalStop()`

Cancels the normal measurement initialization when no save format is selected while auto-save is enabled.

This function is triggered when the user attempts to start a normal measurement with auto-save enabled but closes the format selection dialog without choosing any format (e.g., pressing the X).

It performs the following actions: - Stops any saving process and resets saving state. - Updates UI labels and colors to indicate no measurement is running. - Re-enables the start button and disables pause/stop buttons. - Re-enables all measurement tabs and resumes connection monitoring. - Restores main window tab interactivity.

Returns None

`notSelectedFormatScheduledStop()`

Cancels the scheduled measurement setup when no save format is selected while auto-save is enabled.

This function is triggered when the user attempts to start a scheduled measurement with auto-save enabled but closes the format selection dialog without choosing a format (e.g., presses the close [X] button).

It performs the following actions: - Stops any saving operation and resets saving state. - Updates UI status and color to indicate no measurement is running. - Re-enables the start button and disables pause/stop buttons. - Restores tab interactivity for other measurement modes and the main window. - Resumes the timer for monitoring device connection.

Returns None

`onAutoSaveIntervalChanged()`

Warns the user if they change the auto-save interval from index 0 to any other value.

`pauseLimitedMeasurement()`

Toggles the pause/resume state of a limited measurement based on number of events.

This function allows the user to temporarily pause or resume a limited measurement in progress. It communicates with the background worker thread to suspend or resume data acquisition accordingly.

The function performs the following actions: - If the current button text is “Pause”: - Sends a pause signal to the worker thread. - Updates the button text to “Continue”. - If the current button text is “Continue”: - Temporarily disables the button to avoid duplicate clicks. - Sends a resume signal to the worker thread. - Updates the button text back to “Pause” and re-enables it.

Returns None

`pauseNormalMeasurement()`

Toggles the pause/resume state of a normal measurement.

This function allows the user to pause and later resume an ongoing normal measurement. It interacts with the background worker thread to suspend or resume data acquisition accordingly.

The function performs the following actions: - If the current state is “Pause acquisition”: - Sends a signal to the worker thread to pause the measurement. - Updates the button text to “Continue acquisition”. - If the current state is “Continue acquisition”: - Temporarily disables the button to avoid duplicate clicks. - Sends a signal to the worker thread to resume the measurement. - Restores the button text to “Pause acquisition” and re-enables it.

Returns None

`pauseScheduledMeasurement()`

Toggles the pause/resume state of a scheduled measurement.

This function allows the user to pause or resume an ongoing scheduled measurement. It communicates with the background worker thread to suspend or resume the acquisition process.

The function performs the following actions: - If the current button text is “Pause”: - Sends a pause signal to the worker thread. - Updates the button text to “Continue”. - If the current button text is “Continue”: - Temporarily disables the button to prevent multiple clicks. - Sends a resume signal to the worker thread. - Updates the button text back to “Pause” and re-enables the button.

Returns None

`resetSaveSentinels()`

Resets all save-related sentinels when the save folder is changed.

This ensures that previously saved or autosaved data is marked as outdated, so it can be saved again in the new location if necessary.

Returns None

`resetValuesSentinels()`

Resets all channel sentinels to False.

Returns None

`saveDataAction(format, date)`

Handles the logic for saving measurement data, either from autosave or manual save.

This function determines if the data has already been saved in the requested format. If so, it notifies the user without duplicating the file. If the data was autosaved in a different format, it converts the temporary file to the selected format and saves it. If the data has not been saved yet, it creates a new file with the current measurement data.

Additionally, it displays informational dialogs to confirm the save or alert the user if the process fails.

Parameters

- **format** (*str*) – The desired file format to save (*"txt"*, *"csv"*, or *"dat"*).
- **date** (*datetime*) – The date and time used to generate the filename for saving.

`saveDataButtonAction()`

Opens a dialog for selecting the file format when saving data manually.

This function is triggered when the user clicks the manual "Save" button. It displays a dialog with available formats (*txt*, *csv*, *dat*) and, upon acceptance, calls *saveDataAction* with the selected format and the current date and time.

Returns None

`saveSettings()`

Builds and stores the header for the data file, including device and active channel settings.

The header contains: - Active channels in the measurement. - General device configurations. - Per-channel settings for each active channel.

`setErrorAutoSaving()`

`setValuesBeforeMeasurement()`

Activates channel sentinels based on the selected checkboxes.

Returns None

`settingsAfterMeasurement()`

Restores common UI controls after completing any measurement type.

Re-enables connection, save options, and channel selection checkboxes.

`settingsBeforeMeasurement()`

Applies common UI settings before starting any type of measurement.

Disables save options and channel checkboxes to prevent changes during acquisition.

Returns None

showDialogIncorrectDates()
Shows a warning dialog if the end date is earlier than the start date in a scheduled measurement.

showDialogIncorrectInitDate()
Shows a warning dialog if the start date is earlier than the current date and time.

showDialogNoChannels()
Shows a warning dialog if no channels are selected for measurement.

showProgressDialog(*filename*)
Displays a modal dialog showing the progress of the file saving process.

This function creates and shows a progress dialog with a label and a progress bar. It starts a background worker (*ProcessingDataSaved*) to process the file saving and updates the progress bar in real time via signals. When the process finishes, it triggers the final confirmation message.

Parameters **filename** (*str*) – The full path of the file being saved.

startLimitedMeasurement()
Starts a limited timestamp measurement based on a fixed number of measurement cycles.

This mode runs the acquisition process until the user-defined number of measurements is reached. The function performs the following actions: - Verifies that at least one channel (A, B, C, or D) is selected. - If none are selected, shows an alert and aborts the start. - If valid: - Disables other measurement tabs to prevent mode switching. - Disables the start button and enables pause/stop controls. - Prepares measurement parameters and auto-save options. - Generates the filename if auto-save is active. - Initializes the worker thread in "limited" mode, passing the number of desired measurements. - Connects thread signals to appropriate UI update handlers. - Starts the background thread and updates the UI to reflect the running state.

Returns None

startNormalMeasurement()
Starts a normal timestamp measurement.

Normal measurements begin immediately when the user presses "Start" and continue until manually stopped. This function performs the following actions: - Validates that at least one channel (A, B, C, or D) is selected. - If no channels are selected, an alert dialog is displayed and the measurement is not started. - If valid:

- Disables the start button and enables pause/stop controls.
- Disables other measurement tabs to prevent mode changes mid-measurement.
- Saves pre-measurement settings and prepares auto-save behavior if selected.
- Updates UI labels and status indicators to reflect active measurement.
- Creates and starts a background thread (*WorkerThreadTimeStamping*) to handle the measurement process.
- Connects worker signals to UI update slots.

Returns None

startScheduledMeasurement()
Initiates a scheduled timestamp measurement with start and end times defined by the user.

Scheduled measurements begin at a future time and automatically stop at the configured end time. This function performs the following actions: - Verifies that at least one channel is selected for measurement. - Disables tab navigation to prevent mode switching. - Retrieves and converts the start and end date-time values from the UI fields. - Validates:

- That the start time is not in the past.
- That the end time is after the start time.

- If validation fails, corresponding alert dialogs are shown.
- **If validation passes:**
  - Initializes auto-save format selection if enabled.
  - Configures a countdown timer (*timerBeginMeasurement*) to wait until the scheduled start time.
  - Updates the UI to reflect the waiting state with remaining time.
  - Begins measurement immediately if the scheduled start time is now or has passed.

Returns None

`startTimerConnection()`

Starts the connection monitoring timer to periodically check the device status.

This function activates the *timerConnection*, which verifies the connection with the Tempico device every 500 milliseconds. It is typically called after a measurement has ended or when the device is reconnected, ensuring continuous monitoring of the device's availability.

Returns None

`stopLimitedButtonAction()`

Stops an ongoing limited measurement.

This function disables the pause and stop buttons for the limited measurement mode and sends a stop signal to the worker thread.

Returns None

`stopLimitedMeasurement()`

Stops the limited measurement process, regardless of whether the target number of measurements has been reached.

This function can be used by the user to interrupt the measurement at any point. It performs the following actions: - Marks the measurement as completed (even if prematurely). - Stops the background worker thread. - Updates the UI status and color to reflect that no measurement is running. - Resets the pause button state (if previously in "Continue" mode). - Re-enables the start button and disables pause/stop buttons. - Re-enables the measurement count spinbox for further configuration. - Restores tab navigation for both the local interface and main window. - Signals the main window to re-enable settings and perform cleanup. - Restarts the timer that monitors device connection.

Returns None

`stopNormalButtonAction()`

Stops an ongoing normal measurement.

This function disables the pause and stop buttons for the normal measurement mode and sends a stop signal to the worker thread.

Returns None

`stopNormalMeasurement()`

Stops an ongoing normal measurement and resets the interface to its default state.

This function performs the following actions: - Marks the measurement as completed by setting the corresponding sentinel. - Sends a stop signal to the background worker thread. - Updates the UI to reflect that no measurement is running (label, color, buttons). - If the measurement was paused, resets the pause button

text to "Pause acquisition". - Re-enables tab switching and restores the initial measurement configuration. - Restarts the timer for checking device connection. - Notifies the main window to reflect the updated system state and re-enable settings.

Returns None

`stopScheduledMeasurement()`

Stops a scheduled measurement at any point — before it starts (waiting phase) or during execution.

This function handles both scenarios: - If the measurement hasn't started yet and is waiting for the scheduled start time:

- Cancels the countdown timer and resets all UI elements to their default state.
- **If the measurement is already running:**
  - Stops the data acquisition thread and the timer responsible for countdown to the end.
  - Updates the internal state and UI accordingly.

In both cases, the function: - Resets saving state and UI labels. - Restores tab and button availability. - Notifies the main window to re-enable configuration options. - Calls post-measurement cleanup logic.

Returns None

`stopScyheduledButtonAction()`

Stops an ongoing scheduled measurement.

If the measurement is still in the waiting state, it calls *stopScheduledMeasurement()* directly. Otherwise, it stops the scheduled timer, disables the pause and stop buttons, and sends a stop signal to the worker thread.

Returns None

`stopTimerConnection()`

Stops the connection monitoring timer during an active measurement.

This function is called when a measurement begins to prevent interference from the periodic connection checks. It halts the *timerConnection*, which normally runs in the background to monitor the status of the Tempico device.

Returns None

`updateCheckBoxLabels()`

Updates the visibility of measurement labels and values based on each channel's checkbox state.

This function performs the following actions: - For each channel (A, B, C, D), if its corresponding checkbox is checked: - The label and value indicator showing the number of measurements for that channel are made visible. - If unchecked, both elements are hidden.

This allows dynamic feedback on active channels during timestamp measurements.

Returns None

`updateProgressDialog(percent)`

Updates the progress dialog during the save process.

This function updates the progress bar and label to reflect the current progress percentage. If the progress reaches or exceeds 100%, the dialog is automatically closed.

Parameters `percent` (*int*) – The current progress percentage (0 to 100).

Returns None

updateShowTable()

Toggles the visibility of the timestamp data table based on the checkbox state.

This function performs the following actions: - If the "Show Table" checkbox is checked, the timestamp data table is displayed. - If the checkbox is unchecked, the table is hidden.

Returns None

updateValueTotalMeasurements(*text*)

Updates the label showing the total measurements.

updateValuesMeasurementA(*text*)

Updates the label showing the measurement value for channel A.

updateValuesMeasurementB(*text*)

Updates the label showing the measurement value for channel B.

updateValuesMeasurementC(*text*)

Updates the label showing the measurement value for channel C.

updateValuesMeasurementD(*text*)

Updates the label showing the measurement value for channel D.

CHAPTER

SIX

THREADCOUNTESTIMATED MODULE

class Threads.ThreadCountEstimated.**WorkerThreadCountsEstimated**(*channelASentinel*, *channelBSentinel*, *channelCSentinel*, *channelDSentinel*, *device: pyTempico.core.TempicoDevice*)

Bases: PySide2.QtCore.QThread

This class represents a worker thread that processes Start-Stop measurements in a separate thread to avoid blocking the main GUI thread. It handles data acquisition from the Tempico device, determines the feasibility of performing measurements per channel, and continuously collects and emits measurement data while running.

The thread evaluates which channels meet the minimum requirements for valid measurements (e.g., at least 2 stops within a defined time window), disables those that do not, and allows the user to decide whether to proceed with the remaining ones.

Signals are extensively used to communicate with the GUI, updating labels, graphs, and table data in real time, or notifying when no valid measurements are available.

Additional attributes track the number of stop events per channel and enable synchronization with the main interface using a *threading.Event()* object.

Parameters

- **channelASentinel** – Boolean indicating if Channel A is initially selected for measurement.
- **channelBSentinel** – Boolean indicating if Channel B is initially selected for measurement.
- **channelCSentinel** – Boolean indicating if Channel C is initially selected for measurement.
- **channelDSentinel** – Boolean indicating if Channel D is initially selected for measurement.
- **device** – The TempicoDevice instance used to perform the measurement operations.

Signals:

- newValue(str, float, float): Emitted with updated values for display (label updates).
- updateLabel(str, float, float): Updates the label values for each channel.
- newMeasurement(float, datetime, float, float, float, float, float, float, float, float): Emitted with measurement timestamp, and count/uncertainty data for each channel.
- noTotalMeasurements(): Emitted when none of the channels provide valid stop data.
- noPartialMeasurements(list): Emitted with a list of channels that failed the threshold.
- changeStatusText(str): Updates the system status label with a custom message.
- changeStatusColor(int): Updates the system status label color code.

- disconnectedDevice(): Emitted when the Tempico device is no longer reachable.
- initialDate(str): Emitted with the starting timestamp of the measurement session.
- finalDate(str): Emitted with the ending timestamp of the measurement session.

Attributes:

- channelsMeasure: List of channels accepted for measurement after validation.
- channelsWithoutMeasurements: List of channels that failed the validation criteria.
- continueEvent: Event object used to pause/resume thread logic when user interaction is needed.
- running: Boolean flag to control the thread's main loop.
- numberStopsChannelA/B/C/D: Internal counters to track stop events for each channel.

`calculateIntervalWithStops`(*currentMeasure*, *numberStops*)

Calculates stop time intervals based on pulse timestamps from the measurement.

The function receives a raw measurement and the expected number of stops, then iteratively computes the time differences between consecutive valid pulse timestamps. Each interval is converted to a frequency in Hz by dividing 10^12 by the difference in timestamp units (assuming picoseconds).

Parameters

- `currentMeasure` – A list containing pulse data for a single run.
- `numberStops` – The number of stop intervals expected in the measurement.

Returns A list of frequency values (in Hz) calculated from the stop intervals.

`changeStatusColor`

`changeStatusText`

`determineStopsNumber`(*channelTest*)

Determines the optimal number of stop pulses for a given channel to ensure valid measurements.

This function tests different stop configurations by progressively reducing the number of required stop pulses (from 5 down to 2). For each configuration, it performs multiple test measurements and checks if at least half of them return valid data. If a configuration with sufficient measurements is found, it is returned. If none meet the condition, 1 is returned by default. If the thread is stopped during execution, the process halts.

Parameters `channelTest` – A string indicating the channel to test ("A", "B", "C", or "D").

Returns An integer representing the number of stop pulses that reliably produce measurements.

`disconnectedDevice`

`enableDisableChannels`()

Enables or disables device channels based on active measurement sentinels.

This function disables all channels initially, then enables only the selected ones based on the sentinel flags. It configures each enabled channel with a default stop mask and averaging settings, and adds them to the list of channels to be measured. This reduces processing time by avoiding unnecessary measurements from inactive channels.

Returns None

`finalDate`

getMeasurements()

Retrieves and processes the latest measurement data from the Tempico device.

For each enabled channel, it verifies that the received data is valid, calculates stop intervals based on the number of pulses received, and computes both the mean and standard deviation of the intervals. These values represent the estimated count and its uncertainty.

If valid data is found, the results are emitted via a signal along with the timestamp of the measurement. If no data is available, default values are emitted depending on the state of each channel.

Returns None

initialDate

newMeasurement

newValue

noPartialMeasurements

noTotalMeasurements

run()

Executes the thread's main loop to perform count estimations from the Tempico device.

First, it evaluates each selected channel to determine whether valid measurements can be obtained. If a channel does not meet the required threshold, it is disabled and removed from the measurement process. If at least one channel is valid, the measurement process begins.

The loop runs continuously while the thread is active, triggering measurements every second. If the device becomes unresponsive or the user stops the measurement, the thread exits.

Returns None

staticMetaObject = <PySide2.QtCore.QMetaObject object>

stop()

Stops the measurement process by setting the running sentinel to False.

This function is typically called from the GUI when the user requests to stop the measurement. Once the sentinel is set to False, the running loop in the thread will exit, effectively terminating the background measurement process.

Returns None

updateLabel

CHAPTER

SEVEN

THREADG2 MODULE

class Threads.ThreadG2.**WorkerThreadG2**(*stopChannel: str*, *maximumTime: float*, *numberBins: int*, *coincidenceWindow: float*, *device: pyTempico.core.TempicoDevice*, *mode*, *units*, *limitedMeasurement=False*, *numberOfMeasurements=0*, *autoclearMeasure=False*)

Bases: PySide2.QtCore.QThread

Worker thread for handling g^2 (second-order correlation) measurements in a separate thread to ensure that the GUI remains responsive during data acquisition.

This class is responsible for running the correlation measurement in the background using the Tempico device. It collects start-stop events, builds and normalizes the g^2 histogram, estimates parameters such as counts per second, and communicates results and status updates back to the GUI via signals.

Signals:

- updateMeasurement: Signal emitted to update the g^2 histogram, total starts, and total stops (list, int, int).
- updateTauValues: Signal emitted to update tau values along with the total starts and stops (list, int, int).
- updateStatusLabel: Signal emitted to update the measurement status message in the GUI (str).
- updateColorLabel: Signal emitted to update the color indicator of the status label (int).
- updateEstimatedParameter: Signal emitted to update the current estimated counts per second (str).
- updateDeterminedParameters: Signal emitted when determined parameters are ready to be updated.
- updateFirstParameter: Signal emitted with the first calculated parameter for display (str).

Parameters

- **stopChannel** – The stop channel selected for the measurement ("A", "B", "C", or "D") (str).
- **maximumTime** – The maximum measurement time range (float, in picoseconds).
- **numberBins** – The number of bins used for building the histogram (int).
- **coincidenceWindow** – The coincidence window size (float, in picoseconds).
- **device** – The Tempico device used for data acquisition (tempico.TempicoDevice).
- **mode** – The mode setting that defines measurement configuration (int or str depending on GUI settings).
- **units** – The time units selected for display and processing (str).
- **limitedMeasurement** – Whether the measurement is limited to a fixed number of acquisitions (bool, default=False).

- **numberOfMeasurements** – The number of measurements when in limited mode (int, default=0).
- **autoclearMeasure** – Whether the measurement should periodically auto-clear accumulated histograms (bool, default=False).

buildG2Histogram(*timeDifferences*)

Builds and normalizes the g2 histogram from time differences.

It computes a histogram of the given time differences, accumulates the results with previous data, and normalizes the values using the estimated counts per second, the total integration time, and the coincidence window.

Parameters **timeDifferences** – List of stop time differences from the measurement (list).

Returns Normalized g2 histogram as a NumPy array (ndarray).

changeToOneStop()

Sets the selected stop channel to use only one stop.

Returns None

clearG2()

Resets all g2 measurement values to start a new measurement.

Returns None

estimateParameterInMeasurement()

Estimates differences between stop signals during a measurement.

The method applies the settings for estimated parameters, performs a device measurement, and calculates the time difference between two stop channels when valid data is detected.

Returns A list of time differences between stop signals (list).

estimatedParameterValue()

Estimates an initial parameter value based on device measurements.

The method configures the device for estimation, collects up to 100 measurements, and computes the time differences between stop channels. Progress updates are emitted through *updateStatusLabel* and *updateColorLabel*. If too many measurements are invalid or no start signals are detected, the function returns -1.

Returns Estimated parameter value (float) or -1 if estimation fails.

generateBinList()

Generates the list of histogram bins based on the selected mode and user-defined parameters.

If mode 1 is selected, bins start from 12,500 ps. If mode 2 is selected, bins start from 125,000 ps. The bins extend up to the maximum time with the number of bins specified.

Returns Array of bin edges for the histogram (ndarray).

getCountPerSecondParameter(*estimatedDifferences*)

Calculates the counts per second parameter from stop differences.

It computes the mean of the given stop time differences and converts it into counts per second, assuming values are in picoseconds.

Parameters **estimatedDifferences** – List of stop time differences (list).

Returns Estimated counts per second, rounded to the nearest integer (float).

getDivisionFactor()

Returns the normalization factor based on the selected time unit.

Returns Factor for unit conversion (int).

`getG2Average`(*g2Histogram*)

Calculates the average value of the given g2 histogram.

Parameters `g2Histogram` – Histogram values to average (ndarray or list).

Returns Average value of the histogram (float).

`getMeasurement`()

Performs a measurement cycle to obtain data for g2 analysis.

The method applies measurement settings, collects runs from the device, processes stop events, and builds a g2 histogram from the detected time differences. It also estimates an auxiliary parameter from stop differences and updates status/measurement signals accordingly.

If there are too many missing start events or stops, error messages are emitted. Otherwise, the g2 values and estimated parameters are updated in real time.

Returns None

`processRun`(*run*, *timeDifferences*)

Processes a single run from the Tempico measurement.

Adds stop values to the list of time differences if valid, updates the total number of starts, stops, and the total integration time.

Parameters

- `run` – A single measurement run from the Tempico device (list).
- `timeDifferences` – List where valid stop times are appended (list).

Returns 1 if the stop value is invalid (-1), otherwise 0 (int).

`psToS`(*picoseconds*)

Converts a value from picoseconds (ps) to seconds (s).

Parameters `picoseconds` – Time value in picoseconds (int or float).

Returns Converted value in seconds (float).

`recoverSettings`()

Restores the previously saved configuration to the device.

This re-enables all channels (A–D) and applies the saved values for number of runs, number of stops, stop masks, average cycles, and acquisition modes.

Returns None

`run`(*self*) → None

`saveSettings`()

Saves the current configuration of the device into instance variables.

This includes number of runs, number of stops, stop masks, average cycles, and acquisition modes for each channel (A–D).

Returns None

`settingsForEstimate`()

Configures the device to estimate the photon arrival rate on the selected stop channel.

The function sets the device to perform a single run, disables all channels, and then enables only the selected stop channel (A–D). For the chosen channel, it configures the acquisition mode, stop mask, average cycles, and ensures that two stop events are captured for pulse estimation.

Returns None

settingsForEstimatedParameters()

Applies minimal configuration to the device for parameter estimation.

Unlike the initial setup, this function only sets the number of stops for the selected stop channel (A–D) and configures the acquisition mode. It assumes that the general configuration has already been applied.

Returns None

settingsForMeasurement()

Applies the configuration required to perform the g^2 measurement.

It sets the device to run 100 acquisitions, assigns one stop channel (A–D), and configures its acquisition mode according to *self.modeSettings*.

Returns None

sortByStart(*measurement*)

Sorts the measurement runs based on the start value.

Parameters **measurement** – List of measurement runs (list).

Returns Sorted list of runs by the start value (list).

staticMetaObject = <PySide2.QtCore.QMetaObject object>

stop()

Stops the execution thread by changing the sentinel flag and emitting the signal to terminate all processes related to the thread.

Returns None

updateColorLabel

updateDeterminedParameters

updateEstimatedParameter

updateFirstParameter

updateMeasurement

updateStatusLabel

updateTauValues

CHAPTER

EIGHT

THREADLIFETIME MODULE

class Threads.ThreadLifeTime.**WorkerThreadLifeTime**(*deviceStartChannel*, *deviceStopChannel*, *binwidthText*, *numberMeasurements*, *device*, *TimeRange*)

Bases: PySide2.QtCore.QThread

Worker thread for handling LifeTime (Fluorescence Lifetime Measurement) processing in a separate thread to ensure that the GUI remains responsive during measurements.

This class performs the measurement tasks in the background, processes the data, and communicates with the main thread to update the GUI with the measurement status, results, and other relevant information.

Signals:

- createdSignal: Signal emitted when the worker thread is created.
- statusSignal: Signal emitted to update the status message in the GUI.
- pointSignal: Signal emitted to update the current measurement point.
- updateValues: Signal emitted to update measurement values and corresponding times.
- updateLabel: Signal emitted to update a specific label in the GUI.
- updateMeasurementsLabel: Signal emitted to update the label showing the number of measurements taken.

Parameters

- **deviceStartChannel** – The start channel of the Tempico device used for measurement (TempicoChannel).
- **deviceStopChannel** – The stop channel of the Tempico device used for measurement (TempicoChannel).
- **binwidthText** – The selected bin width for the measurement (str).
- **numberMeasurements** – The number of measurements to be taken (int).
- **device** – The Tempico device used for performing measurements (Tempico).
- **TimeRange** – The time range for the measurements (int).

checkDeviceStatus()

Checks the status of the device by attempting to read a parameter.

If an error occurs during the read operation, the measurement process is stopped.

Returns None

clear()

Clears the recorded start-stop differences and resets the total measurement count.

This function empties the list that stores the differences between start and stop measurements, and sets the total number of measurements back to zero.

Returns None

createLifeTimeData()

Generates LifeTime data from the start-stop differences and emits the calculated results to the main thread.

Processes the collected data to determine the number of counts within a given time based on the maximum value of start-stop differences. Normalizes the data according to the determined units and calculates the histogram counts.

Returns None

createdSignal

disconectedSignal

getBinWidthNumber()

Calculates the bin width in picoseconds based on user input in different units.

Extracts the numerical value and its corresponding unit from the user-provided text, and converts it into picoseconds for further calculations.

Returns None

getUnits(*picosecondsValue*)

Determines the appropriate units for a given value in picoseconds.

Converts a given value in picoseconds to the most suitable time unit (picoseconds, nanoseconds, microseconds, or milliseconds).

Parameters **picosecondsValue** – The value in picoseconds (float).

Returns A list containing the unit as a string and the factor by which to divide the value (list).

measurementMode()

Sets the measurement mode for the device channels based on the specified time range.

If the time range is less than or equal to 500000 ps , sets all channels to mode 1. Otherwise, sets all channels to mode 2.

Parameters **self** – The instance of the class.

Returns None

pointSignal

run()

Runs the measurement thread, continuously taking measurements based on user-selected values and passing them to the graphs.

Returns None

staticMetaObject = <PySide2.QtCore.QMetaObject object>

statusSignal

stop()

Stops the measurement process by setting the running flag to False.

Emits a status message indicating that the measurement is ending and updates the status bar color to yellow.

Returns None

`takeMeasurements`(*percentage*)

Takes measurements from the device and updates the start-stop differences list.

Configures the device for measurement runs and checks the input and stop channels for data. If no measurements are detected, emits a status signal to update the main thread about the current measurement state.

Parameters `percentage` – The percentage of measurement completion (float).

Returns None

`updateLabel`

`updateMeasurementsLabel`

`updateValues`

CHAPTER

NINE

THREADPROCCESSDATA MODULE

class Threads.ThreadProccessData.**ProcessingDataSaved**(*filename*)

Bases: PySide2.QtCore.QThread

Thread class responsible for processing saved measurement files when autosave is not enabled.

This class runs in a separate thread to avoid blocking the main UI while processing data from a previously saved file. It emits progress updates so the user interface can reflect the current status of the processing task.

Main responsibilities: - Read and process the specified saved data file. - Emit progress updates during the processing. - Operate independently from the main thread to keep the UI responsive.

Signals:

changeProgress (float): Emitted to update the processing progress in the UI as a percentage.

Parameters **filename** – Path to the saved measurement file to be processed.

changeProgress

run(*self*) → None

sortTimeStamps(*file_path*)

Sorts a saved timestamp file by start times and emits progress updates during processing.

This function reads the specified measurement file, parses its data entries, sorts them chronologically by start time, and writes the sorted content back to the same file. Progress updates are emitted throughout the process to inform the UI of the current completion percentage.

Parameters **file_path** – Path to the saved measurement file to be sorted.

Returns None

staticMetaObject = <PySide2.QtCore.QMetaObject object>

CHAPTER

TEN

THREADSTARTSTOP MODULE

class Threads.ThreadStartStop.**WorkerThreadStartStopHistogram**(*parent*, *device: pyTempico.core.TempicoDevice*, *sentinelSaveA*, *sentinelSaveB*, *sentinelSaveC*, *sentinelSaveD*, *checkA*, *checkB*, *checkC*, *checkD*)

Bases: PySide2.QtCore.QThread

Worker thread for processing Start-Stop measurements without blocking the main UI.

This class extends *QThread* to handle the acquisition and processing of Start-Stop measurement data from a *TempicoDevice* in the background. It collects data from up to four channels (A–D), applies range validation, detects out-of-range measurements, and emits signals to update the GUI with processed results, status messages, and visual indicators.

Main responsibilities: - Acquire Start-Stop measurements for each enabled channel. - Track total and out-of-range counts per channel. - Manage sentinels for detecting missing measurements or channels requiring mode changes. - Maintain and emit maximum measurement values for updating histogram ranges. - Save device configuration before measurement and determine if batch mode should be used. - Handle dialog prompts when channels exceed acceptable out-of-range limits.

Signals:

dataSignal (float, str): Emitted with the processed measurement value and its channel ID.

dataPureSignal (float, str): Emitted with the raw measurement value and its channel ID.

threadCreated (int): Emitted when the thread is initialized and ready.

dialogInit (): Emitted to initialize a measurement-related dialog in the UI.

colorValue (int): Emitted to update the UI status color (e.g., normal, warning, error).

stringValue (str): Emitted to update the UI status text.

dialogSignal (str): Emitted to request opening a dialog for a specific channel.

newMaxValueSignal (float, str): Emitted when a new maximum value is detected for a channel.

dataBashSignal (list, list, list, list): Emitted with batched processed measurements for all channels.

dataPureBashSignal (list, list, list, list): Emitted with batched raw measurements for all channels.

Parameters

- **parent** – The parent widget or window managing this thread.
- **device** – Instance of *tempico.TempicoDevice* used to acquire measurements.
- **sentinelSaveA** – Boolean indicating if channel A measurements should be saved.

- **sentinelSaveB** – Boolean indicating if channel B measurements should be saved.
- **sentinelSaveC** – Boolean indicating if channel C measurements should be saved.
- **sentinelSaveD** – Boolean indicating if channel D measurements should be saved.
- **checkA** – Boolean indicating if channel A should be checked for measurement validity.
- **checkB** – Boolean indicating if channel B should be checked for measurement validity.
- **checkC** – Boolean indicating if channel C should be checked for measurement validity.
- **checkD** – Boolean indicating if channel D should be checked for measurement validity.

addChannelWarning(*channel*)

Adds the specified channel to the list of channels that need to change mode.

Parameters **channel** – The channel to be added to the warning list (str).

Returns None

applyCurrentSettings()

Applies the stored configuration settings to the device.

This function sends the previously saved general and per-channel configuration parameters back to the connected device. These include run counts, thresholds, average cycles, modes, number of stops, stop edge types, and stop masks for each channel. After applying the settings, it enables or disables the channels according to the stored configuration.

Returns None

changeMaxValue(*channel*, *valueMax*)

Changes the maximum value for the specified channel.

Parameters

- **channel** – The channel for which the maximum value is to be changed (str).
- **valueMax** – The new maximum value to be set (float).

Returns None

checkCorruptData(*currentMeasurement*, *nextMeasurement*)

Checks if a measurement is corrupt by comparing it to the next measurement.

The function verifies whether the sequence number of the next measurement is either the same or exactly one greater than the current one. If it is neither, the current measurement is considered corrupt and should be skipped.

Parameters

- **currentMeasurement** – Current measurement data.
- **nextMeasurement** – Next measurement data to compare against.

Returns 1 if the current measurement is corrupt, 0 otherwise.

colorValue

compareMaxValue(*newValue*, *channel*)

Checks if a new measurement value exceeds the current maximum for the given channel.

If the new value is greater than the stored maximum for that channel, the maximum is updated and a signal is emitted to notify that the histogram's X-axis range should be adjusted.

Parameters

- **newValue** – The new measurement value to compare.
- **channel** – The channel identifier ("A", "B", "C", or "D").

Returns None

dataBashSignal

dataPureBashSignal

dataPureSignal

dataSignal

dialogInit

dialogIsOpen()

Sets the 'openDialog' sentinel to False, indicating that no dialog is open.

Returns None

dialogSignal

emitOrSaveMeasurement(*listValues*, *measurement*, *pureMeasurement*, *pureList*, *channel*)

Determines whether a measurement should be saved or emitted, filtering out values that exceed the device's maximum limit.

If the measurement is below the predefined limit, it is either stored for batch processing or emitted immediately depending on the operating mode. In batched mode, the measurement and its raw value are appended to the provided lists. In normal mode, the measurement is compared against the current maximum value and emitted through the corresponding signals.

Parameters

- **listValues** – List to store processed measurement values in batched mode.
- **measurement** – Processed measurement value.
- **pureMeasurement** – Raw measurement value from the device.
- **pureList** – List to store raw measurement values in batched mode.
- **channel** – Channel identifier for the measurement.

Returns None

enableDisableChannels()

getMeasurements()

Acquires and processes timestamp measurements from each active channel.

This function retrieves measurement data from the connected device, validates it, and processes it channel by channel (A–D). It checks the number of stops for each channel, determines whether to emit the measurement immediately or wait based on the configured number of runs, and detects potential corruption in the measurement sequence.

Additional checks are performed to: - Identify channels that are not recording measurements correctly. - Detect if the problem is due to missing Start signals. - Prompt the user to change the mode of channels if a high proportion of

out-of-range values is detected.

- **Capture potential disconnection errors and stop the measurement if they** persist.

This method also emits status messages, channel problem notifications, and measurement values for both normal and batched data modes.

Returns None

`getRange`(*currentMeasurement*, *stopNumber*)

Determines the valid range of stops to process for a measurement.

This function calculates the maximum valid index range based on the expected number of stops and the actual length of the measurement data. It ensures that the processing does not exceed available data, which helps prevent errors in cases where measurements are incomplete or corrupted.

Parameters

- `currentMeasurement` – Current measurement data as a list.
- `stopNumber` – Expected number of stops for the measurement.

Returns The validated number of stops to process.

`isBashedMeasurement`()

Determines whether measurements should be processed in batches.

If the configured number of runs is greater than 50, the measurement mode is set to batched; otherwise, it is set to normal.

Returns None

`newMaxValueSignal`

`run`()

Executes the thread's main loop to update the graph based on measurements.

The function emits a signal indicating that the thread has been created. It then enters a loop that continues as long as the thread is running. Inside the loop, it calls the update method to refresh the graph and pauses for 0.5 seconds before the next iteration.

Returns None

`saveCurrentSettings`()

Saves the current configuration settings of the device.

This function retrieves the general and per-channel configuration parameters from the connected device and stores them in the corresponding instance variables. These values include run counts, thresholds, modes, number of stops, stop edge types, and stop masks for each channel.

Returns None

`staticMetaObject = <PySide2.QtCore.QMetaObject object>`

`stop`()

Stops the measurement process and closes the thread.

This method emits a signal to indicate that the thread has been stopped and sets the running flag to False.

Returns None

`stringValue`

`threadCreated`

CHAPTER

ELEVEN

THREADTIMESTAMPING MODULE

class Threads.ThreadTimeStamping.**WorkerThreadTimeStamping**(*channelASentinel*, *channelBSentinel*, *channelCSentinel*, *channelDSentinel*, *normalMeasurementSentinel*, *scheduleMeasurementSentinel*, *limitMeasurementSentinel*, *device: pyTempico.core.TempicoDevice*, *savefile*, *filename*, *isAutoSave*, *maximumMeasurements=0*)

Bases: `PySide2.QtCore.QThread`

Thread class responsible for executing and managing timestamp measurements in the background.

This class extends *QThread* to handle real-time acquisition of timestamp data from a *TempicoDevice* without blocking the main UI thread. It supports three operational modes: - **Normal**: Manual start/stop by the user. - **Scheduled**: Automatic start/stop based on configured start and end times. - **Limited**: Stops automatically after a predefined number of measurements.

The class collects data from up to four independent channels (A–D), tracks total and per-channel counts, and emits signals to update the user interface with measurement data, status text, and status colors. It also handles autosaving of data to files in supported formats.

Main responsibilities: - Continuously acquire timestamp measurements while the thread is running. - Track total measurements and per-channel statistics. - Handle sentinel flags for pausing, stopping, and controlling measurement flow. - Emit signals to update UI components with real-time measurement and status information. - Save measurement data automatically or manually based on user settings. - Enable or disable specific channels based on configuration.

Signals:

newMeasurement (tuple, tuple, tuple, tuple, tuple, float, float, float, float, float): Emitted when new measurement data is available, including per-channel and total counts.

changeStatusText (str): Emitted to update the UI status text.

changeStatusColor (int): Emitted to update the UI status color (e.g., ready, running, saving).

finishedMeasurements (): Emitted when the measurement process is complete.

Parameters

- `channelASentinel` – Flag to enable/disable channel A measurements.
- `channelBSentinel` – Flag to enable/disable channel B measurements.
- `channelCSentinel` – Flag to enable/disable channel C measurements.
- `channelDSentinel` – Flag to enable/disable channel D measurements.

- **normalMeasurementSentinel** – Flag to control normal mode execution.
- **scheduleMeasurementSentinel** – Flag to control scheduled mode execution.
- **limitMeasurementSentinel** – Flag to control limited mode execution.
- **device** – Instance of *tempico.TempicoDevice* used to acquire timestamp data.
- **savefile** – Object responsible for handling file-saving operations.
- **filename** – File path or name to save measurement data.
- **isAutoSave** – Boolean indicating whether autosave is enabled.
- **maximumMeasurements** – Maximum number of measurements in limited mode (default 0).

applyCurrentSettings()

Applies the previously saved device configuration.

This function restores the general and per-channel configuration parameters that were saved from the device, and reconfigures the device accordingly. It also enables or disables channels based on sentinel values to continue measurements.

Returns None

changeIsPauseFalse()

Sets the pause sentinel to False.

Returns None

changeIsPauseTrue()

Sets the pause sentinel to True.

Returns None

changeReadyToReorder()

Sets the ready-to-reorder sentinel to True.

Returns None

changeStatusColor

changeStatusText

checkCorruptData(*currentMeasurement*, *nextMeasurement*)

Checks if the current measurement should be skipped due to data corruption.

This function compares the sequence index of the current measurement with the next one. If the sequence does not follow the expected order, it determines that the current measurement is corrupt and should be skipped.

Parameters

- **currentMeasurement** – The current measurement data as a list.
- **nextMeasurement** – The following measurement data as a list.

Returns 1 if the current measurement should be skipped, 0 otherwise.

detectedErrorSaving

enableChannelsAfterFinishedMeasurement()

Enables all measurement channels on the device.

enableDisableChannels()

Configures the device channels according to the user's selection.

This function enables or disables each measurement channel (A–D) based on the sentinel values provided at initialization. It also retrieves the number of stops for each active channel and calculates the total possible measurements for a single run, storing it in *totalDataPerMeasurement*.

`finishedMeasurements`

`getLimitedMeasurements()`

Acquires, processes, and validates measurements from the device in limited mode, stopping once the configured maximum number of measurements is reached.

This method: 1. Solicits a measurement from the device and captures any console output. 2. Handles timeouts by attempting additional fetches, aborting when data is stagnant, or resetting the device after repeated failures. 3. Sorts the retrieved measurements by start time when applicable. 4. Iterates through the measurements and: - Detects and skips corrupt data sequences. - Assigns valid measurement values to their respective channels (A, B, C, D) based

on enabled channel sentinels.

- Verifies that values fall within the allowed measurement range.
- Updates total and per-channel measurement counters.
- Stops processing immediately if the maximum limit is reached.

5. Identifies active channels producing no valid data and emits user warnings.
6. Detects and records start-only measurements (with no corresponding stop values).

7. Emits the processed results to the main thread, including: - Tuples of *(start_time, value)* for each channel. - List of start-only measurement timestamps. - Per-channel totals and overall total. 8. Sends progress updates and status messages to the GUI.

If no measurements are available or a start channel is inactive, a warning is emitted to the main thread.

`getMeasurement()`

Acquires, processes, and validates measurements from the device, assigning them to their corresponding channels and emitting results to the main thread.

This method: 1. Solicits a measurement from the device. 2. Handles timeouts by attempting additional fetches and aborting if needed. 3. Verifies measurement integrity: - Detects and skips corrupt data sequences. - Confirms that the device returns complete measurements. 4. Organizes measurements by channel (A, B, C, D) according to the channel sentinels enabled in the worker. 5. Validates that each enabled channel is actually producing measurements, generating user warnings if not. 6. Detects and reports cases where only start values are recorded without associated stop measurements. 7. Updates internal counters for total and per-channel measurements. 8. Emits the processed measurement data, including: - Tuples for each channel with *(start_time, value)*. - List of start-only measurements. - Totals per channel and global total.

If no measurements are available or a start channel is inactive, a warning is emitted to the main thread.

`getRange(`*currentMeasurement*, *stopNumber*`)`

Determines the valid range of stops in a measurement.

This function checks the length of the provided measurement list to ensure it is not corrupted. If the length is smaller than expected, the range is adjusted to avoid out-of-bounds errors when processing the data.

Parameters

- `currentMeasurement` – The current measurement data as a list.
- `stopNumber` – The expected number of stops.

Returns The validated number of stops to iterate over.

`newMeasurement`

`run`(*self*) → None

`saveCurrentMeasurements()`

Saves the current device configuration settings.

This function retrieves the general and per-channel configuration parameters from the connected device and stores them in the corresponding instance variables. These values represent the active measurement settings, such as run counts, thresholds, modes, and stop configurations.

Returns None

`sortMeasurementByStart`(*measurement*)

Sorts measurement records by their start time.

Filters out empty runs from the provided measurement list and sorts the remaining entries in ascending order based on the third element of each record (start time).

Parameters `measurement` (*list*) – List of measurement records, where each record is an iterable containing at least three elements.

Returns A list of filtered and sorted measurement records.

Return type list

`sortTimeStamps`(*filePath*)

Sorts a timestamp file by start times and moves the processed data to the user-selected folder.

This function reads a temporary autosave file containing timestamp data, sorts the entries by their start time, and writes the sorted data both back to the temporary file and to the final file in the user-selected save location. The function also updates the processing progress via status signals.

Parameters `file_path` – The destination file path where the sorted data will be saved.

Returns None

`staticMetaObject = <PySide2.QtCore.QMetaObject object>`

`stop()`

Sets the running sentinel to False to indicate that the thread has finished.

Returns None

`syncTime()`

Synchronizes the system time with the connected device.

CHAPTER

TWELVE

FINDDEVICES MODULE

class Utils.findDevices.**PyTempicoManager**

Bases: `object`

Class for managing the search and connection of Tempico devices connected to serial ports on the PC.

This class is responsible for scanning the available serial ports on the system and searching for Tempico devices connected to those ports. It facilitates interaction with Tempico devices by allowing their identification and connection.

Parameters **None** – No parameters are required for the initialization of the class.

Returns None

find_devices(*vid_s*, *pid_s*)

Finds and verifies whether a device with the given VID and PID is a Tempico device.

This function takes the Vendor ID (VID) and Product ID (PID) as inputs, converts them to integers, and attempts to open the device using these values. It then checks if the manufacturer and product strings match the expected values for a Tempico device.

Parameters

- **vid_s** (*str*) – The Vendor ID (VID) of the device in string format.
- **pid_s** (*str*) – The Product ID (PID) of the device in string format.

Returns *True* if the device is a Tempico, *False* otherwise.

Return type bool

get_pytempico_devices()

Searches for Tempico devices among the available serial ports and returns a list of their port names.

This function scans the system's serial ports, checks if they correspond to a Tempico device by retrieving the VID and PID, and verifies if they match the Tempico device specifications. Bluetooth devices are excluded from the search.

Returns A list of port names corresponding to Tempico devices.

Return type list of str

get_vid_pid(*vid_pid_information*)

Extracts the Vendor ID (VID) and Product ID (PID) from a string and returns them as a tuple.

This function processes a string that contains the VID and PID information in the format 'VID:PID=xxxx:yyyy'. It splits the string and retrieves the VID and PID values, returning them as a tuple of strings.

Parameters **vid_pid_information** (*str*) – A string containing the VID and PID information.

Returns A tuple containing the VID and PID as strings (vid, pid).

Return type tuple

`verify_pyTempico`(*tuple_vid_pid*)

Verifies whether the connected device is a Tempico device.

This function checks if the device's Vendor ID (VID) and Product ID (PID) match the values corresponding to a Tempico device. It returns *True* if the device is identified as a Tempico, and *False* otherwise.

Parameters **tuple_vid_pid** (*tuple*) – A tuple containing the VID and PID of the device.

Returns *True* if the device is a Tempico, *False* otherwise.

Return type bool

CHAPTER

THIRTEEN

MAIN MODULE

class main.**MainWindow**(*parent=None*, **args*)

Bases: PySide2.QtWidgets.QMainWindow

Main application window class.

This class is responsible for creating the main window of the application, including its tabs and graphical user interface (GUI) components. It serves as a bridge between the UI classes that handle the design elements and the logic classes that manage the application's functionalities.

The main tasks of this class include: - Initializing the main window and its layout. - Creating and managing tabs for different sections of the application. - Integrating UI design elements with the logical operations for each functionality.

Parameters

- **parent** (*QWidget, optional*) – The parent widget (optional).
- **args** (*tuple*) – Additional arguments.

Helpg2Button()

about_settings()

Opens the About window displaying information about the company, version, and repository.

This function opens a window showing details about the company (Tausand), the software version, and the location of the repository. It does not perform any checks or validations and directly displays the About window.

It does not take any parameters and does not return a value. :returns: None

activeMeasurement()

Activates the sentinel indicating that a measurement is currently running.

This function sets the internal flag to True, which is used to track the active state of a measurement process.

Returns None

clicked_tabs()

Executes an action when a tab is clicked and creates the corresponding window.

This function checks which tab is currently active and constructs the associated window by invoking the appropriate function. If the tab corresponds to the Start/Stop Histogram, it stops the LifeTime timer and constructs the Start/Stop Histogram window. If the tab corresponds to the Lifetime Measurements, it constructs the Lifetime window and sets up the LifeTime logic if it has not been initialized yet.

It does not take any parameters and does not return a value. :returns: None

closeEvent(*event*)

Handles the close event of the main window and prompts the user for confirmation.

This function is triggered when the user attempts to close the main window. It displays a dialog asking if the user is sure about closing the application (Tempico software). If the user confirms by selecting "Yes", the event is accepted and the application closes. If the user selects "No", the close event is ignored, and the application remains open.

Parameters **event** (*QCloseEvent*) – The close event triggered when attempting to close the window.

Returns None

construct_counts_estimated(*parent*)

Constructs the Counts Estimated window.

This function takes a *QTabWidget* parent, and if the sentinel is not set, it creates an instance of the *Ui_CountsEstimated* class and sets up the UI using the given parent.

It ensures the UI is initialized only once by checking the *sentinel3* flag.

Parameters **parent** (*QWidget*) – The parent widget (typically a *QTabWidget*) for the counts estimated window.

Returns None

construct_g2(*parent*)

Constructs the g2 Measurements window.

This function takes a *QTabWidget* parent, and if the sentinel is not set, it creates an instance of the *Uig2* class and sets up the UI using the given parent.

It does not return a value.

Parameters **parent** (*QWidget*) – The parent widget (typically a *QTabWidget*) for the lifetime measurements window.

Returns None

construct_lifetime(*parent*)

Constructs the Lifetime Measurements window.

This function takes a *QTabWidget* parent, and if the sentinel is not set, it creates an instance of the *UiLifeTime* class and sets up the UI using the given parent.

It does not return a value.

Parameters **parent** (*QWidget*) – The parent widget (typically a *QTabWidget*) for the lifetime measurements window.

Returns None

construct_start_stop_histogram(*parent*)

Constructs the Start/Stop Histogram window.

This function takes a *QTabWidget* parent, and if the sentinel is not set, it creates an instance of the *Ui_HistogramaStartStop* class and sets up the UI using the given parent.

It does not return a value.

Parameters **parent** (*QWidget*) – The parent widget (typically a *QTabWidget*) for the histogram window.

Returns None

construct_time_stamping(*parent*)

Constructs the Counts Estimated window.

This function takes a *QTabWidget* parent, and if the sentinel is not set, it creates an instance of the *Ui_CountsEstimated* class and sets up the UI using the given parent.

It ensures the UI is initialized only once by checking the *sentinel3* flag.

Parameters **parent** (*QWidget*) – The parent widget (typically a *QTabWidget*) for the counts estimated window.

Returns None

disconnect_button_click()

Handles the disconnect button click event.

This function hides the graphical display, disables the disconnect button, and re-enables the connect button. It also closes the connected device and resets its reference to *None*. If additional graphics like *g2Graphic* or *LifeTimeGraphic* are active, it will disconnect them as well.

It does not take any parameters and does not return a value. :returns: None

disconnectedDevice()

Handles device disconnection without displaying any user interface dialogs.

This function is triggered when the connection with the device is lost. It notifies all relevant components to handle the disconnection and sets the connected device reference to None.

Returns None

enableSettings()

Re-enables the device and general settings dialogs if they are open.

If either the device configuration or general settings dialog is currently visible, this function refreshes and reactivates their interactive fields by calling the corresponding *get_settings()* and *enableSettings()* methods.

Returns None

folderPrefixClicked()

general_settings_clicked()

Opens the general settings window for device-wide configurations.

This function checks if a device is connected. If no measurement is currently running, it opens the general settings window where configurations that affect the entire device (regardless of the channel) can be adjusted. If a measurement is running, a message box is displayed to inform the user that changes cannot be made while the measurement is in progress. If no device is connected, a message box is shown indicating that no device was found.

It does not take any parameters and does not return a value. :returns: None

manageConection()

Manages the connection status with the measurement device.

This function checks if the device is still connected. If the connection is lost, it stops the timer responsible for monitoring the device, removes the device instance, disables measurement functionalities in all tabs, and displays an error dialog to inform the user.

Returns None

noMeasurement()

Deactivates the sentinel indicating that no measurement is currently running.

This function sets the internal flag to False, marking that there is no active measurement process at the moment.

Returns None

open_dialog()

Opens a dialog window to detect and connect a measurement device.

This function creates and displays a dialog window that lists available measurement devices and their corresponding ports. Users can select a device and either connect or cancel the operation.

If the 'Connect' button is clicked: - The selected device is connected, and all relevant software options are enabled. - The function tries to open the device and, upon success, interacts with graphical components (if any are present) to reflect the connected status.

If the 'Cancel' button is clicked: - No options are activated.

In case of a connection error, a message box is displayed informing the user of the failure.

Returns None

parameters_action()

resetSaveSentinelsAllWindows()

resizeEvent(*event*)

Handles the window resize event and adjusts the column widths of the parameters table.

This function is triggered when the user resizes the main window. It resizes the columns of the parameters table in the LifeTime tab based on the current window width. The column widths are scaled proportionally to ensure the table adapts to the new window size.

Parameters **event** (*QResizeEvent*) – The resize event triggered when the window is resized.

Returns None

saveSettings()

Retrieves and stores the current device configuration into internal variables.

This function queries the connected device for channel-specific settings such as averaging cycles, mode, number of stops, stop edge type, stop mask, threshold voltage, and number of runs. These values are saved into instance attributes for later use, such as restoring settings or displaying them in the UI.

Returns None

settings_clicked()

Opens the settings window when the settings option is clicked.

This function checks if a device is connected. If no measurement is currently running, it displays the settings window for channel configuration. If a measurement is running, a message box is shown informing the user that changes cannot be made while a measurement is in progress. If no device is connected, a message box alerts the user that no device was found.

It does not take any parameters and does not return a value. :returns: None

staticMetaObject = <PySide2.QtCore.QMetaObject object>

class main.**SplashScreen**

Bases: PySide2.QtWidgets.QMainWindow

Splash screen class for the application.

This class creates a splash screen that displays an image for a brief period before fading out and transitioning to the main application window.

The key responsibilities of this class include: - Displaying a splash screen with a specified image. - Timing the display duration and handling the transition to the main application window.

Parameters **parent** (*QWidget, optional*) – The parent widget (optional).

show_main_window()

Displays the main window of the application.

This function creates an instance of the main window (*MainWindow*) and displays it. After showing the main window, it closes the current window.

It does not take any parameters and does not return a value. :returns: None

staticMetaObject = <PySide2.QtCore.QMetaObject object>

main.**execProgram()**

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